

Event Density: A Unified Framework for Physics

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Availability: the corpus, this report, and the certified-simulation scripts are public at github.com/allen-proxmire/ED-generative (archived, all-versions DOI 10.5281/zenodo.20149496); the substrate simulator and companion evaluation scripts live in the event-density working repo.

Abstract

Event Density (ED) is a substrate ontology whose unifying power comes from one primitive, the **arrow of time** — the irreversible act of commitment — doing load-bearing work across sectors that mainstream physics treats as separate and even as in tension. The same arrow selects the quantum pointer basis (einselection) and *is* gravity’s preferred time (the khronon), so quantum measurement’s “which basis is definite” and gravity’s “which time is flowing” are one primitive wearing two hats, and the problem of time dissolves rather than being reconciled. On that spine, ED is **form-complete across every box a unified framework must check and value-inherited exactly where the Standard Model’s free parameters live** — a structural consequence of a pure-relation ontology having no intrinsic scalar. It is not a gap, and no worse than the Standard Model’s own standing on those numbers.

What “form-complete” means here is concrete: in each box, the substrate fixes the *form* of a law while the universe supplies its *numbers*. Gravity is the cleanest case. ED’s gravity is khronometric — the arrow made dynamical — and it reproduces the weak-field Einstein metric and the classical tests, with $c_T = c$ structural and $\alpha_2 = 0$ (the latter shared with GR, a survival rather than a distinction). The same khronon field gives the galactic dark-matter phenomenology (the MOND deep field) and *specifies but does not yet deliver* a single- Λ cosmology: the CMB and clusters are an open, live falsification risk, not a passed fit. Its lone scale is inherited through H_0 , exactly as the Standard Model inherits its parameters. The scorecard in §2 records the same verdict box by box, at an explicit tier for each — quantum mechanics reconstructed, charge native, chirality a theorem, mass from binding — with the section that earns each row.

This report is as forthright about its limits as its results. Two structural questions are genuinely open — the gauge multiplicities $\{1, 2, 3\}$ and whether exact charge conservation forces anomaly-freedom on the emergent chiral content — and they reduce to two residuals of one form-complete reduction (the substrate-to-Dirac de-

scent) — #1 to the channel-topology wall, #2 to a sibling residual, not the same wall — plus one conjectured route to it; both are unsolved in the Standard Model too. One wall is *proven*: the finite-memory substrate cannot reach the primality escape layer. The constants are inherited by design. And, decisively, **no distinctive, argument-ending prediction is confirmed yet**: ED makes one confirmed forward prediction (the mobility-law shape) and survives the tests that killed rival gravity theories ($c_{\text{GW}} = c$), while its flagship distinctive bet, that the MOND scale evolves as $a_0(z) = cH(z)/(2\pi)$, has met its first data with a partial result — the evolution is seen and a constant a_0 excluded decisively (ED’s qualitative call confirmed, MOND’s picture killed), but ED’s exact rate is mildly disfavored by that one survey. The bets that would single ED out cleanly remain open (§15). One primitive recurring as the mechanism across these sectors — most strikingly at the black hole, where the arrow’s irreversibility preserves information as an un-erasable record — is a serious case for a unified framework; it is not yet the empirical case that would end the argument.

§1 — The Claim and the Bar

The claim

Event Density is a substrate ontology — commitment and participation, with the arrow of time as the single process primitive — advanced here as a **unified framework for physics**. The claim is not that ED reproduces one sector well. It is that one primitive does load-bearing work across sectors that mainstream physics treats as separate: quantum measurement, gravity, gauge structure, charge, chirality, and mass. This report’s goal is to state, box by box and at an explicit tier, how much of that is earned and how much is open.

Two words on what Event Density is *not*. It is not a claim to have solved physics — the report carries two genuine structural opens, one proven wall, and no confirmed distinctive prediction, and it says so in every relevant section. The framework also avoids the labels “theory of everything” and “grand unified theory”: the first overclaims, the second technically excludes gravity, and both trigger dismissal for reasons unrelated to content. “A unified framework for physics” is the accurate description and the ceiling of the claim.

The boxes a unified framework must check

A framework earns the word “unified” by accounting for a specific list, and this report is organized around it:

- a minimal foundation (§3);
- quantum mechanics from that foundation (§4);
- entanglement, Bell and Tsirelson (§4);
- gravity — the weak field, the khronometric class, galactic dark matter, and dark energy (§5);
- cosmology and the dark sector — inflation, and the CMB dark-matter frontier (§7);
- quantum mechanics and gravity unified rather than glued (§6);
- the gauge forces $SU(3) \times SU(2) \times U(1)$ (§8);
- charge quantization (§9);
- electromagnetism (§9);
- chirality and parity violation (§10);
- the matter content — spinors, masses (§11);
- the constants (§12);
- internal consistency, i.e. anomalies (§13);
- and, the scientific test, novel falsifiable predictions (§15).

A framework that checks these boxes with derivations, and is explicit about where it inherits or stops, is a unified framework in the sense this report defends.

The bar

The evidential standard is deliberately high, and the report’s credibility depends on holding it. Extraordinary claims require extraordinary evidence, and for a unified-framework claim from outside the mainstream, the

operative form of that standard is this: **the limits must be as loud as the results**. This document states each result and each open at its true tier, and marks the one thing that is proven impossible.

Concretely, three things are kept rigorously distinct throughout:

1. **What is derived** — a result that follows from the substrate (at the tiers below).
2. **What is inherited** — a value ED's own logic says it cannot and need not set (the constants, §12), on the same footing as the Standard Model's free parameters.
3. **What is walled or open** — a limit, either proven (the one primality wall, §14) or a genuine open derivation. The report refers to the two open derivations by number throughout: **#1** is the *representation spectrum* — why the gauge multiplicities are exactly $\{1, 2, 3\}$ (§8) — and **#2** is whether the substrate's exact charge conservation *forces* anomaly-freedom on the emergent matter (§13). Both are stated as open, not smoothed.

The tiers

Every substantive claim in the report is tagged with one of these tiers, taken from the corpus's own discipline. They are the vocabulary of the whole document, and no claim is stated above its source's tier.

- **Derived** — follows from the primitives or the substrate structure (a proof or forced construction).
- **Form-forced / derived-conditional** — the *form* is forced given stated inputs (e.g. the gauge group given the complex amplitude); the conditions are disclosed.
- **Reconstructed** — substantially grounded by mapping onto a standard theorem, with the theorem's rigor assumed (the quantum-logic keystone, §4).
- **Structural / account** — a grounded argument or identification, not a proof; the unification move (§6) is the report's central account-tier claim.
- **Measured** — a result produced by running the *certified* substrate simulator: a direct simulation of the primitives whose dynamics are validated against the substrate's own invariants and conservation laws, so its outputs are measurements of ED itself rather than of a stand-in model (the binding-mass mechanism, §11; the charge skeleton, §9).
- **Grounded** — a primitive supplies the operational content, conditionally or structurally.
- **Inherited** — a value or structure taken from measurement or standard mathematics by ED's own logic (the constants, e.g. the cosmological scale H_0 ; the Lorentzian metric signature; Piron–Solèr).
- **Wall** — a proven or acknowledged limit, the order ED does not reach (primality is the one *proven* wall).

The tier word carries the honesty. A result marked “reconstructed” or “account” is not a weaker result dressed up; it is a result stated at exactly the confidence it has earned. The scorecard that follows (§2) tags every box this way, and the sections deliver on each tag.

The scorecard's Tier column draws on this vocabulary, often **compounded** — a box can be derived in *form* and inherited in *value* (gravity), or grounded on one front and gated on another (anomalies) — and sometimes with a one-word note on *how* a result was reached: **analytic** (a pen-and-paper derivation, as opposed to *measured* on the simulator), **synthesis** (an organizing claim spanning several primitives, as at the minimal foundation), **gated** (an open derivation blocked on a named reduction, e.g. #2 on the substrate-to-Dirac descent), and **falsifier frontier** (the predictions box, which is empirical rather than a derivation tier). No cell sits above its source's tier.

One aggregate verdict recurs, built from these tiers: **form-complete** means the substrate fixes the *form* of the law — the equation, structure, or mechanism — in every box a unified framework must check, at one of the derivation tiers above, even where the law's numerical values are inherited. It is deliberately weaker than *closed*, which would claim nothing is left open; the report claims form-completeness, not closure.

The primitives it leans on

The substrate's postulates are thirteen (P01–P13, canonical in foundations/Paper_087) plus two memory kernels (V1, V5). The report refers to these by number; the ones it leans on are:

- **P02 — participation:** the primitive relation, chains participating on channels. The “participation” half of the foundation.
- **P03 — channel/locus indexing:** channels and loci are indexed, and space is homogeneous (no preferred place).
- **P04 — bandwidth:** each participation carries a non-negative, additive scalar b_K , the “amount of participation.”
- **P05 — polarity-transport:** a connection-like structure carries polarity along graph edges, the substrate root of gauge content.
- **P06 — spatial dimension:** space is three-dimensional (plus one time, via P13); postulated, not derived.
- **P07 — channel structure:** channels are ontologically distinct carriers, distinguishable even at equal bandwidth and polarity.
- **P09 — $U(1)$ polarity:** each participation carries an angular phase $\pi_K \in U(1)$, the substrate’s primitive phase.
- **P11 — commitment irreversibility:** at a commitment event a chain’s multi-channel participation collapses, irreversibly, to a single channel. **This is the arrow of time** — the report’s one load-bearing process primitive.
- **V1 — the single-chain kernel:** the retarded memory rule by which a chain’s prior participation propagates to its present (Paper_089).
- **V5 — the cross-chain kernel:** the finite-reach retarded coupling that correlates separate chains (Paper_090), the substrate source of composite systems and entanglement.

(P01, P08, P10, P12, P13 complete the set; the substrate scale P08 is empirically identified with the Planck length ℓ_P , an inherited value, not a derivation.)

§2 — The Scorecard

The report’s structure is this table. Read down the “arrow’s job” column and the thirteen boxes are not thirteen separate achievements — they are one primitive, the arrow of time, showing up in twelve of them (every box but the constants, the value layer it does not set). Read the “tier” column and the tier is explicit in every row. Every  is delivered by its section at the stated tier; every  is a genuine open carried openly; every — is a principled inheritance, not a hole.

Box a unified framework must check	ED status	The arrow’s job here	Tier	§
Minimal foundation	 commitment + participation; one process does every job	the arrow <i>is</i> the sole process primitive	Grounded / synthesis	§3
Quantum mechanics from the substrate	 reconstructed (orthogonality reduced, $\mathbb{C}$ selected, Born non-contextuality grounded; residual = Solèr lattice rigor)	the arrow selects the pointer basis (einselection) + supplies repeatability	Reconstructed	§4

Box a unified framework must check	ED status	The arrow's job here	Tier	§
Entanglement (Bell / Tsirelson)	✓ closed arc (non-factorizability, monogamy, no-signaling forced, the Bell-Tsirelson bound); read as the pre-individuation regime, not action-at-a-distance	commitment (P11) individuates; entanglement is the not-yet-committed regime	Closed (Grounded/Derived) + interpretation	§4
Gravity (GR + galactic MOND + dark energy)	✓ GR quartet closed (GR-I..IV) + galactic MOND passes (KM-I); ! dark energy/ Λ : form identified, magnitude inherited via H_0 (§12) — the CMB/cluster dark sector is §7	the khronon <i>is</i> the arrow made dynamical (preferred time)	Derived (weak field) / grounded (galactic); values inherited	§5
Cosmology and the dark sector	✓ inflation = the substrate saturating (consilience); ✓ galactic MOND read as a horizon-interference cross-term (native); ! the CMB/cluster dark component = a native <i>candidate</i> (a committed neutral relic), not a fit — the sector's live falsification risk	the arrow saturates (inflation) + interferes with the horizon (MOND)	Consilience (inflation) / grounded (interference) / candidate-not-delivered (CMB); values inherited	§7

Box a unified framework must check	ED status	The arrow's job here	Tier	§
QM + gravity unified	✓ distinctive — the same arrow is pointer basis <i>and</i> preferred time (the problem of time dissolves); and the black-hole information paradox is addressed by a substrate-level unitarity-preserving mechanism (information = the arrow's record; no Hilbert-space unitarity proof claimed)	one arrow, two demands	Account (distinctive)	§6
Gauge forces SU(3)×SU(2)×U(1)	✓ shape derived (SU(N) form, non-abelian, F^2 , gap, single U(1)); ! {1,2,3} the wall	the single hypercharge = the arrow wearing a phase	Derived-conditional / Wall (#1)	§8
Charge quantization	✓ native (integer winding, integral Gauss law)	irreversibility quantizes + protects the winding	Measured	§9
Electromagnetism (smooth field)	✓ coherence-weighted limit is Coulomb; the field is a coarse-grained shadow	the arrow sorts the continuum (field = shadow)	Analytic + structural	§9
Chirality / parity violation	✓ operator grounded; theorem: clean substrate vector $\forall N \Rightarrow$ parity violation necessarily spontaneous; ! casting inherited	the arrow is half of γ^5 ; parity-cleanness forces the vector theorem	Derived (theorem) + inherited casting	§10

Box a unified framework must check	ED status	The arrow's job here	Tier	§
Matter content (spinor, mass)	✓ substrate-to-Dirac form-complete; native matter/antimatter (C); native binding-mass ("mass without mass"); mass ≠ time dilation; ! generations/Higgs inherited	the arrow undoubles the spinor + selects matter over antimatter (C) + makes the lone front massless + sets the clock	Form-complete + measured (mass)	§11
The constants (α, mass ratios, H_0)	— inherited by design — no intrinsic scalar (the A1 zero-capacity result, §12); the SM inherits them too	— (the value layer the arrow does not set)	Inherited-by-design	§12
Internal consistency (anomalies)	✓ conservation + clean-vector baseline solid; ! chiral cancellation inherited; one candidate gated on the substrate-to-Dirac reduction's residual (the channel-topology wall; the reduction itself form-complete, §11)	clean-vector baseline (arrow) + the worldline-arrow spectral flow	Grounded baseline / gated (#2)	§13

Box a unified framework must check	ED status	The arrow's job here	Tier	§
Novel falsifiable predictions	✓ ranked weapons + one confirmed forward prediction (the degenerate- mobility law, §15); the flagship distinctive bet ($a_o(z) =$ $ch(z)/(2\pi)$) has met first data — evolution seen, constant a_o excluded decisively (ED's call confirmed, MOND's killed), exact rate mildly disfavored (data run faster than $H(z)$; $\sim 1-2\sigma$ with systematics); ! no argument-ending weapon fully confirmed	several sharp falsifiers trace to the arrow ($a_o(z)$, $w=-1$, the clean-vector prohibitions)	Falsifier frontier	§15

How to read the three marks

- ✓ — ED delivers this, at the tier in the tier column. A parenthetical residual that is inherited standard mathematics (Piron-Solèr) or a rigor-completion does not demote the check; the tier word carries it (e.g. “re-constructed,” not “proven”).
- — — inherited by design. ED's own logic (no intrinsic scalar) says it cannot and need not set this. Not a hole, and no worse than the Standard Model's standing on the same numbers.
- ! — a genuine open. There are exactly two structural ones: the gauge multiplicities #1 (§8) and the chiral anomaly cancellation #2 (§13), which reduce to two residuals of one form-complete reduction (the substrate-to-Dirac descent) — #1 to the channel-topology wall, #2 to a sibling residual — plus one conjectured route (the linking argument for #1). Additionally, the state of the predictions: no argument-ending weapon is fully confirmed yet, though the flagship $a_o(z)$ bet has a live partial confrontation (§15).

This scorecard is the skeleton of the report; the arrow column is its thesis; the tier column is its conscience. The sections deliver each row.

§3 — The Arrow of Time: the one process that does every job

The inversion

The microscopic laws are time-reversal invariant; the arrow of time is imposed on them from outside, by a low-entropy boundary condition. That is the standard settlement, and it makes irreversibility a derived, statistical, almost apologetic fact, something the fundamental description does not itself contain.

Event Density inverts this. The arrow is not derived; it is the substrate's single primitive act. The dynamics are based on one-way commitment, noted as P11, and everything reversible in the continuum limit is what survives coarse-graining, not what was there to begin with. This is the opposite of the usual order of explanation, and it is worth being explicit about the cost: ED's substrate carries *less* symmetry than its own continuum limit, and time-reversal invariance is an emergent artifact rather than a fundamental input.

The payoff is the claim of this report: **sector after sector of physics, worked independently, lands back on that one commitment**. The arrow is not an entry on ED's list of results. It is the mechanism doing the work in most of them.

What the arrow *is* in ED

In ED the substrate is not a field or a particle or a piece of space. It is a set of relations, chains participating on channels, each carrying an amplitude, and the one thing that happens to those relations is **commitment**: a participation resolves, irreversibly, one way and not the other. That act is the primitive ED calls P11, and it is the arrow of time. Not a clock, not a coordinate, not the ticking of a background parameter. The arrow is the commitment itself: the substrate deciding, once, without taking it back.

Two words carry the whole ontology. **Commitment and participation**. Participation is what is there to be resolved, a chain, on a channel, here, now. Commitment is the resolving. Everything else in this report is built out of that pair, and time is not a third thing added to them. Time *is* the commitment. That is why ED does not need a separate temporal stage: space provides a bare index the substrate sits on, but time is not a stage at all, it is the act.

The claim

It is tempting to say “the arrow is the only primitive”. The precise version is sharper and survives scrutiny. When you audit ED's primitives one by one against commitment, they sort cleanly into three piles (the reduction audit of the minimal-ontology paper):

- The things the arrow **acts on** (participation, bandwidth, channels, polarity), which it presupposes and cannot be, because you need something there to commit.
- The **arena** it acts in — the bare substrate index (adjacency, dimension, grain scale) and the available rule-types, with the metric emergent on top rather than given — which carries no direction of its own.
- The arrow **and every process built from it**: the causal chaining of events, the transport that connects them, the descent that drives them, all forward, all the arrow.

So the precise claim is not that commitment is the only thing. It is that **commitment is the only process**. It is the one source of anything happening at all. A single source of process is exactly what you would need for a unified theory. If everything that *happens* traces to one act, then the sectors of physics are not independent. They are that one act, wearing different clothes.

The jobs — one arrow, sector after sector

What follows is the backbone of the report. Each line is a job the arrow does, and each job has a full section below.

Time and irreversibility [this section]. The face-value job. The arrow is the direction, the “now” that commits and does not uncommit. Everything else is downstream of this.

The quantum pointer basis [§4]. Ask any interpretation of quantum mechanics its hardest question, why measurements come out definite in *this* basis and not some other, and you get silence or hand-waving. In ED the answer is the arrow. A commitment can only resolve in the basis it commits along, the channel basis, so the definite outcomes are definite *because* the arrow selected that basis when it committed. Einselection, which every other account has to install by hand or argue for from decoherence, is simply what the arrow is. ED is a definite-outcome theory because it is a committing theory.

The covering law, the geometry of quantum logic [→ §4]. Once a channel is committed it is locked; ask it again and you get the same answer. That locking is irreversibility, the arrow again, and it is exactly the repeatability that the quantum-logic reconstruction needs to stand its Hilbert space up. The arrow does not just pick the basis; it supplies the reason measurement is repeatable at all.

Gravity's preferred time [§5]. ED's gravity is khronometric: it has a preferred foliation of spacetime, a "which time is really flowing" that general relativity insists cannot exist. In ED it not only exists, it is forced, because the arrow *is* a preferred direction of time, and a preferred direction is a preferred foliation. The khronon field that carries ED's gravity is the arrow, spread across the substrate.

The unification move [§6]. Hold those two jobs next to each other. The pointer basis of quantum mechanics and the preferred time of gravity are not two facts. They are the same arrow. That single identity is ED's answer to quantum gravity, and it is the heart of this report: the two theories were never separate to begin with, they share a spine, and that spine is time committing.

Undoubling the spinor [§11]. Put a Dirac field on a lattice and it doubles: you get sixteen species where nature has one. The standard fixes are technical and cost you a symmetry. ED's substrate is a lattice, and the arrow undoubles it down to a single species, because it is a genuine direction that breaks the symmetry the doublers rely on. This is not free: the arrow pays the Nielsen–Ninomiya price by being *non-Hermitian* (its retardation is the Wilson term), which is exactly the premise the no-go forces you to surrender. The electron is one electron because the substrate commits.

Matter's very existence [§11]. The one chirality-adjacent thing ED gives natively is the matter/antimatter reference, the charge-conjugation label C. It comes from the arrow: first arrival, first commitment, selects one side. Matter is here rather than annihilated because something committed first.

Parity [§10]. The handedness operator of the Standard Model, γ^5 , factors in ED into two pieces, the arrow times a spontaneous choice of orientation. The arrow supplies half of chirality directly. (Which force actually *uses* that handedness is inherited representation theory, and §10 is careful about that line.)

The single hypercharge [§8]. The Standard Model has exactly one gauged U(1), one hypercharge, and no one can say why one. In ED there is one because there is one arrow to phase against. Hypercharge is the substrate's polarity measured against the single external flow, and the single external flow is time committing.

Sorting the continuum [§3, §9]. Take any smooth law of physics and ask ED which class it falls in, structure-making, structure-erasing, or structure-preserving. The sorter is the arrow. It is what tells Maxwell from diffusion from a conservation law, because the distinction is entirely about how each one treats direction.

Ten jobs, and they are not ten coincidences. They are the pattern you would expect if commitment really is the only process: apparently unrelated structures turning out to be different faces of one act.

Two guardrails

Two guardrails, stated here so the rest of the report can lean on them.

First, this is a **convergence, not a theorem**. No single derivation takes commitment and produces all ten jobs from it. Each is worked separately, each landing back on the arrow. That is consilience; and it is the kind of evidence unification is actually made of, one mechanism across the sectors. But it is not a proof that the arrow is logically prior to everything, and this document will not pretend it is. A skeptic is right to press here, and §6 presses back: the force of the claim is the recurrence, not a reduction.

Second, the arrow is **not everything** — but the "arena" is narrower than it sounds. What ED takes as primitive is only the *bare* substrate index: its adjacency, its dimension (P06), its grain scale (P08), and the available rule-types. It is **not** "space" in the usual sense. The metric geometry, physical distance, and even the boundary between one system and another are *emergent*, built by commitment: the weak-field metric comes out of the connectivity graph with no lengths on its edges (§5), and a spatial boundary is the coarse-grained shadow of an individuation surface (§4). So space is largely a *result*, not a stage; what stays primitive is the bare index, and even its dimension has a conjectured route home (the linking argument, §14). A process cannot hand you the bare index it runs on, and the later inherited numbers (the constants, the representation spectrum) trace

partly to that same fact: the arrow is a process, not a stage. Naming that limit precisely is what keeps the spine load-bearing instead of mystical.

What this section buys the rest of the report

From here on, no sector is a standalone result. When §4 reconstructs quantum mechanics, watch for the arrow selecting the basis. When §5 closes gravity, watch for the arrow as the preferred time. When §6 unifies them, it is not introducing a new idea, it is pointing out that the basis-selector and the preferred-time are the same object you met here. The Standard Model sectors, §8 through §11, each open by naming which job of the arrow they are.

The boxes a unified theory must check are the skeleton of this report. The arrow of time is the blood that runs through it. Everything after this page is the same one-way act, seen from one more angle.

Related papers:

- *Commitment and Participation: the Minimal Ontology*
- *The Arrow Sorts the Continuum*
- *The Thirteen Primitives
- Position papers

§4 — Quantum Mechanics from the Substrate: the arrow selects the pointer basis

The arrow's job here. Of the roles set out in §3, this section addresses two: the arrow as the quantum **pointer basis** (einselection) and as the source of measurement **repeatability** (the covering law). The claim is not that ED is *compatible* with quantum mechanics. It is that quantum mechanics' most under-explained structural facts, why outcomes are definite in one basis and not another, and why measurement is repeatable at all, are what a one-way commitment *is*.

The postulate that would not ground

ED's quantum kinematics (Paper_004) rests on an inner product over the substrate states — the single load-bearing postulate the whole quantum sector hangs on. Grounding that inner product is what this report calls **the quantum-logic keystone**: get it, and Hilbert space, the Born rule, and the rest of quantum kinematics follow; miss it, and the quantum sector is postulated rather than reconstructed. In a substrate ontology the inner product cannot simply be assumed; it must trace to the primitives or be flagged as postulated. Two assumptions had blocked three prior attempts: that distinct channels are orthogonal, and that the probability assignment is the Gleason/Born one. The obstruction was always the same shape. Orthogonality was sought as a *metric* fact about vectors, and no ED primitive delivers a metric.

The reconstruction breaks the stall by changing the question, from “are the channel vectors orthogonal?” to “what does the substrate operationally do, and what representation must carry that?” The target is the standard quantum-logic chain: **Piron** (an irreducible, complete, atomistic, orthomodular lattice with the covering law, rank ≥ 4 , is the lattice of closed subspaces of a Hermitian space over a division ring with involution), then **Solèr** (an infinite orthonormal sequence of equal norm forces the field into $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ and the space to be a genuine Hilbert space), then a physics input to select the field. ED's relational-lattice is the set of propositions “the commitment resolves in channel-set S ” — the P11 outcomes. The reconstruction maps each axiom to a primitive, with an explicit tier.

Three results

1. Orthogonality reduces to distinguishability. Take candidate channel-states with a *tunable* overlap $c = \langle K|L \rangle$ (do not set $c = 0$). The best confusion-free detection of K , maximizing $\langle K|E|K \rangle$ over POVM elements

subject to $\langle L | E | L \rangle = 0$, is the projector onto the complement of $|L\rangle$:

$$\max \langle K | E | K \rangle = 1 - c^2,$$

which equals 1 if $c = 0$. Perfect distinguishability $\iff$ orthogonality. And ED's channels *are* perfectly distinguishable from commitment frequencies alone (a pure channel-L state has $b_K = 0$, so it commits to K with frequency zero, P07). So $\langle K | L \rangle = 0$ is not an independent postulate; it is what “perfectly distinguishable by commitment” means. The boundary is this: the POVM argument runs *inside* the inner-product representation, so it reduces orthogonality to distinguishability *given* that the representation exists — it does not build the metric from primitives.

2. The complex field is selected. Solèr allows $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$; ED picks $\mathbb{C}$ by two standard discriminators, each grounded in a primitive. $\mathbb{R}$ is excluded not because a real space cannot carry a phase (it can, via a Stueckelberg J), but because a real division ring supplies no *central scalar* square root of -1 ; ED's phase $e^{i\pi}$ is a scalar amplitude factor (Paper_001), not an operator constrained to commute with all observables, so the field must contain i . $\mathbb{H}$ is excluded by the standard tensor-product obstruction: composite systems (built from **V5** — ED's cross-chain coupling, the finite-reach retarded kernel that correlates separate chains, Paper_090; the composite amplitudes are Paper_063) need a commutative field for $\otimes$ to be well-defined. This is a *selection* from Solèr's menu, account-tier, not a from-scratch derivation, but it means $\mathbb{C}$ is no longer a bare assumption.

3. The covering law: irreversibility enforces it. The covering law's operational content is the ideal first-kind measurement, measure an atom, get “yes,” project, and get “yes” again on repetition. The natural worry is that P11's irreversibility *breaks* the repeatability this needs. It is exactly backwards. Commitment projects the state onto the resolved channel-atom (orthogonally, by P11 exclusivity), and then *locks* it: an irreversible outcome cannot drift, so re-measurement returns the same channel with certainty. **A committed channel is the most repeatable outcome there is.** Irreversibility is not an obstruction to the covering law; it is what supplies it. (Scope: this grounds the first-kind core on the *channel* basis; the full lattice-geometric exchange condition and the phase basis remain residual.)

The two arrow-tied resolutions

Two structural questions that other accounts install by hand resolve here directly from the arrow, and they are ED's distinctive quantum picture.

Einselection is primitive. Commitment, P11, is the only collapse primitive, and it is a *which-channel* selection. A superposition $|+\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$ is not an intrinsic channel; no primitive collapses to it; committing a $|+\rangle$ state resolves it to $|1\rangle$ or $|2\rangle$, never to $|+\rangle$. So the channel basis is the unique pointer basis, selected by the arrow, not emergent from environmental decoherence. Phase stays a genuine coherence observable (interference is real; a definite phase and a definite channel are complementary), but it is never a commitment basis. **ED is a preferred-basis quantum theory because it is a committing theory.** Ordinary basis-democracy is recovered at the apparatus level (an apparatus is a system whose channels couple to the target observable), so standard QM emerges; the distinctive claim is that einselection is fundamental rather than derived.

Born probabilities are non-contextual. The bandwidth b_K is intrinsic (P04, a bare state property with no apparatus argument), so $P(K) = b_K / \Sigma b$ is fixed. Because ED commits only in the channel basis, there is exactly *one* substrate commitment context, and Kochen–Specker contextuality (which needs one projector sitting in multiple incompatible contexts) has nowhere to arise. ED assigns non-contextual *probabilities* (Gleason-permitted, the Born rule), never non-contextual definite *values* (Kochen–Specker-forbidden), because outcomes are stochastically committed, not pre-valued. It is consistent with both theorems at once.

Status

This is a **coherent reconstruction, not a closed theorem**. Precisely:

- Orthogonality is *reduced* (to distinguishability, given the representation), not produced from primitives; $\mathbb{C}$ is *selected*, not derived; the covering law is *candidate-grounded* on the channel basis only.

- The residual is **Solèr lattice rigor**, and it is a *different kind of thing* from ED's structural opens (§14). The postulate that actually stalled three prior attempts — channel-orthogonality — is **reduced to operational channel-distinctness given the representation** (Move 1, circularity-audited): not independently closed, but no longer a separate blocking axiom. What remains is a rigor-completion of two parts. Most of it is the standard Piron–Solèr machinery, which ED *invokes* rather than owes — it sits on the “inherited by design” list alongside the lattice-gauge dictionary. The one genuinely ED-side soft spot, named precisely, is the **Solèr exchange-property geometry**: the exchange condition Piron's projective step needs to build the vector-space representation, together with the infinite-dimensional rigor. That technical package rides underneath the whole reconstruction — including under the §4 orthogonality reduction, which is proven inside that representation. Orthomodularity itself is not the open screw: it rides with the representation, holding automatically for any inner-product lattice. This is why the box is checked at the tier the word carries: **reconstructed** — substantially grounded by mapping onto a standard theorem, not proven from scratch, and not a missing physics result of the #1/#2 class.
- ED's quantum *logic* (the orthomodular lattice, the covering law) is substrate-exact; its quantum *geometry* (the inner-product metric) is Born-statistical and therefore emergent. The Hilbert space is exact-as-logic, emergent-as-metric.
- The equal-norm (angle) condition is operationally motivated in the amplitude representation $\langle K|K \rangle = b_K$, so equal-norm $\iff$ equal-bandwidth given P03 homogeneity; the keystone paper carries this as a *candidate*, with an equal-norm circularity that remains a residual.

What is new and solid, independent of the residual rigor, is the **reframe**: orthogonality is operational rather than metric, the complex field is selected by ED's actual phase and composites, and the arrow selects the pointer basis.

Entanglement: the pre-individuation regime

ED has a **closed entanglement arc** (Papers 063–070): composite states are non-factorizable (the tensor-product structure), the Schmidt decomposition follows, monogamy comes from the finite cross-chain bandwidth budget, no-signaling is **forced and over-determined**, von Neumann entropy is grounded, and the **Bell–Tsirelson polytope** ($Bell \subsetneq Tsirelson \subsetneq NS$, placing ED exactly at the quantum ceiling with PR-boxes excluded) is *form-forced* — conditional on V5 enforcing a Hilbert-space inner-product structure (Paper_069) — while its numerical value $2\sqrt{2}$ is inherited as standard mathematics, not derived from primitives. All of it is carried by **V5**, the cross-chain kernel: the finite-reach correlation between distinct chains. (Tier: the arc is closed; no-signaling is Derived-forced; the Tsirelson polytope is form-forced conditional on the V5–Hilbert postulate, its $2\sqrt{2}$ value inherited.)

The interpretation is where the arrow returns. Entanglement, in ED, is the **unresolved regime of participation-rule individuation** (Paper_072): a configuration where two would-be systems share cross-chain participation and **no commitment has yet individuated them** into distinct identities. Two entangled particles are not two things exchanging a spooky influence; they are **one un-individuated chain-complex expressed at two end-points**. Measurement is P11 commitment completing the individuation, which is why entanglement is consumed when it is measured, and why no-signaling is not a constraint ED must impose but one it over-determines: there is no second, separate system for a signal to reach.

This is also where ED's account of space does work. In the substrate, **individuation is prior to spatial separation**: the boundary between two systems is the coarse-grained *shadow* of an individuation surface, so a pre-individuated pair has no such boundary, which is exactly why entanglement reads as “non-local.” The apparent distance between the endpoints is an emergent-metric quantity (§5), and the pre-individuated pair is not yet committed to that metric. (Honest scope: the chains still occupy substrate loci; what is undefined before commitment is the *emergent* separation, not location itself. Making that bridge fully explicit — entanglement as **pre-spatial** — is a named open synthesis, not yet a written result.)

Finally, the same cross-chain kernel that carries entanglement also carries the black hole's interior structure, so ED exhibits an **ER=EPR-class echo** (Paper_071): entanglement and horizon geometry are governed by the one

V5 object. This is a *structural echo*, not a constructive derivation of ER=EPR. (Tier: Grounded interpretation for the pre-individuation reading; a structural echo, not a theorem, for the ER=EPR link.)

What this buys

Note what did the work in every result above: the *irreversibility* of commitment. It locked the channel to give repeatability; it made the channel basis the unique pointer basis; it made outcomes stochastic commitments rather than pre-valued facts. That irreversibility is the arrow of §3. Hold onto the specific identification, **the arrow selects the quantum pointer basis** — because §5 will show the same arrow fixes gravity’s preferred time, and §6 will argue those are not two facts.

Related papers.

- *The Quantum-Logic Keystone: Gleason Reconstruction*
- *Gleason-Type Uniqueness*
- *Pre-Individuation Amplitudes*
- Entanglement arc: *Tensor Product through Bipartite Synthesis*
- *Bell-Tsirelson reconstruction*
- *Individuation Regime*
- *ER=EPR echo*

§5 — Gravity: the khronon is the arrow made dynamical

The arrow’s job here. In §4 the arrow selected the quantum pointer basis. Here it does something a relativist is trained to say is forbidden: it supplies gravity with a **preferred time**. General relativity insists no such thing exists, that no foliation of spacetime is physically singled out. ED’s gravity has one, and it is not smuggled in, it is the arrow of time from §3 made dynamical. The field that carries ED’s gravity, the **khronon**, is the substrate’s direction of commitment, promoted to a degree of freedom. Gravity, in ED, is what the arrow looks like at large scale.

This is the sector where ED is most complete and most distinctive. It reproduces general relativity where GR is tested, and it departs from GR exactly where the arrow leaves a fingerprint.

The weak field is Einstein’s

The lapse of the emergent metric is set by the substrate bandwidth, $N^2 \sim b$, and carrying that through with the correct factor of two yields the **weak-field Einstein metric** and the three classical tests (perihelion precession, light bending, Shapiro delay) at the standard values (GR-I, Derived). Newton’s constant comes out as $G = c^3 \ell_P^2 / \hbar$ once the substrate grain is identified with the Planck length (Paper_027; ℓ_P inherited, the relation derived). So at the level everyone actually measures, ED gravity is the textbook metric. Nothing exotic is on offer in the regime where GR is confirmed, which is the correct behavior for a theory that has to survive the Solar System.

The class is forced, and it is khronometric

What kind of gravity is it, structurally? Lovelock’s theorem plus a mode count fixes the answer: ED gravity is in the **khronometric class**, two tensor modes plus one scalar khronon (GR-II, Derived/Grounded). The tensor sector propagates at the speed of light, $c_T = c$, *structurally* rather than by fine-tuning, so ED is clean against the GW170817 bound that killed a swathe of modified-gravity models. The Lorentz violation the khronon carries is universal and confined to the safe sector.

The dynamical rule for the khronon field is $\ddot{b} = D\nabla^2 b - \kappa\rho$, built and run as a simulation (GR-III, Derived/Measured): it has the Newtonian fixed point, gives $r_s \propto M$, produces a frozen $b \rightarrow 0$ horizon with an area law, and fixes the khronon speed $c_s = c$. And the preferred-frame parameters, the usual place a

preferred-time theory dies, are safe: $\alpha_2 = 0$ **exactly**, and $\alpha_1 = -4\lambda_{\text{local}}$ with $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$, safe by at least seventy orders of magnitude because commitment is sparse (GR-IV). The exact $\alpha_2 = 0$ is not a fitted escape; ED gets it structurally, without tuning, because both gravitational cones are luminal. It is a sharp *kill-switch* (any measured $\alpha_2 \neq 0$ falsifies ED), but GR-IV is explicit that it is a general feature of any luminal-cone khronometric theory, shared with general relativity, so it is a survival ED passes cleanly, not a result that distinguishes it from GR. The distinctive gravitational departure is the MOND deep field, below.

One field: gravity, dark matter, dark energy

The reason to care about this sector is not that it re-derives GR. It is that the *same* khronon does three jobs that mainstream physics assigns to three unrelated things.

- In its **high-acceleration** face, it is Newton and the Einstein metric (above).
- In its **deep field**, it is the dark-matter phenomenology. KM-I embeds MOND relativistically as the khronon's low-acceleration behavior, forced given the ED combination rule $a = \sqrt{a_N \cdot a_0}$ (Paper_030), with the transition scale $a_0 = cH_0/(2\pi)$ coming from the cosmic decoupling surface (Paper_029), the slope-4 Baryonic Tully–Fisher relation $v^4 = GMa_0$ with zero intrinsic scatter (Paper_031), and the AQUAL field equation (Paper_036). Lensing and PPN pass, and there is no new ghost (the w -sector adds no time derivatives); but KM-I is explicit that its feedback into the kinetic matrix is the same regulator degeneracy the cosmological sector below leaves unfit, so the full strong-coupling stability is conditional, not a closed proof (KM-I, Grounded/conditional). ED does not add dark matter; it reads the rotation curves off the same field that gives Newton.
- In its **cosmological** face, it is the dark-fluid and a single cosmological constant Λ . ED *identifies* Λ 's form (the V1 kernel integrated over the cosmic horizon; the smallness is reframed as the Friedmann ratio $(H_0/M_P)^2$, so the *naïve* mode-tower version of the 122-order catastrophe does not arise) and inherits its *magnitude* through H_0 (KM-II, Paper_038_5; the residual hierarchy reduces to the open Route-A derivation of H_0). See §12.

That is the one-line summary of this section, and it is worth stating: **gravity, galactic dark matter, and dark energy are one field in ED, and that field is the arrow.** (Paper_OneField_Letter is the capstone; §7 develops the cosmological dark sector and is explicit that the CMB dark component is a *separate* question from this one field.)

Where ED departs from GR (the falsifiable edge)

These are not caveats, these are the sector's weapons:

- **ED gravity is MOND, not Einstein, in the deep field**, and the emergent metric is kinematic rather than a dynamical Einstein field (the nonlinear completion is the interference cross-term, not the Einstein–Hilbert term). This is a genuine departure, and it is the source of the dark-matter phenomenology rather than an add-on.
- $\alpha_2 = 0$ **exactly** — a sharp kill-switch ED passes structurally, though shared with GR (a survival, not a departure); any measured $\alpha_2 \neq 0$ falsifies ED.
- **The MOND scale evolves with redshift**, $a_0(z) = cH(z)/(2\pi)$. Because a_0 is tied to the cosmic horizon $R_H = c/H$, it tracks the instantaneous Hubble rate rather than being a constant of nature, so it grows with redshift ($\times 1.8$ by $z = 1$). The power $\alpha = 1$ is forced not by dimensional analysis (ED has a second scale, ℓ_P , so a bent power is dimensionally legal) but by the projection mechanism plus ED's no-intrinsic-scalar result (§12), which leaves no dimensionless number free to bend it at the fixed magnitude, so ED cannot fit a_0 the way MOND does. It is the sector's most distinctive live weapon: first data (MUSE-DARK III, 2026) confirm a_0 evolves and exclude a constant a_0 decisively, matching ED's local value to $\sim 8\%$, while mildly disfavoring ED's exact rate (the observed evolution runs somewhat faster than $H(z)$, a few- σ tension from one first-generation survey). This is the flagship of §15.
- **Why a_0 evolves but Λ does not, on the one horizon.** a_0 and Λ are tied to the *same* cosmic horizon ($R_H = c/H$, above), yet one evolves and one is fixed, and the asymmetry is *forced, not chosen*. a_0 rides the horizon *live*, at each epoch's expansion rate, so it evolves. Λ cannot: a vacuum energy tracking the *live*

horizon scales as $\rho_\Lambda \propto H^2$, which gives $w = 0$ and no acceleration at all. So Λ must be the horizon's *frozen* saturation-floor value, which is why $w = -1$ exactly. a_0 is a scale and can ride the live rate harmlessly; Λ is a vacuum energy and cannot. (One residual open: whether the substrate *cleanly forces* the frozen-floor value or *stipulates* it; Paper_038_5 §3.5.)

- **Weak-lensing depends on substrate activity** (Paper_038.6), and the **merging-cluster offset-velocity “knee”** is a distinctive shape prediction (Paper_OffsetVelocityLaw), though its sharp leg is a survey-era test, not experiment-ready today (§15). These are on the falsifier frontier (§15).

The debts

It is a closed sector, but it still has edges:

- **Clusters and the CMB are the sector's live falsification risk** — no CMB spectrum is computed and no cluster profile is fit, and this is where MOND-class theories characteristically die. The full cosmological dark sector, its candidate component, and this failure mode are developed in §7.
- **Λ 's magnitude is inherited via H_0** (its form is identified and the naïve mode-tower catastrophe dissolved — §12; the number reduces to H_0 , whose substrate derivation, Route A, is the open piece).
- **GW polarization content** beyond $c_T = c$ is viable but its exact couplings are not fully derived, the one live tension, not a failure.

The GR quartet is closed and Derived, and the galactic MOND phenomenology (KM-I) passes its lensing and PPN kill-checks with no new ghost (though its full strong-coupling stability ties, conditionally, to the same regulator degeneracy below). But the cosmological half is not delivered: the one-field dark-matter/dark-energy unification is a *structure specified with an unfixed dial*, and whether it survives the CMB and clusters is open and could kill the sector. So the sector is closed where GR is tested and genuinely at risk where this theory class historically fails; “one field” should not be read as “one field that has passed cosmology.”

What this buys

The load-bearing fact to carry forward is the identification made at the top: **gravity's preferred time is the arrow**. The khronon is not a new field standing next to the arrow; it is the arrow, dynamical. Hold that against §4's result, *the arrow selects the quantum pointer basis*, and §6 has both halves it needs. The pointer basis and the preferred foliation are jobs of one primitive. That is the unification, and gravity is where its second half lives.

Related papers.

- *Weak-Field Einstein Metric, Khronometric Class, Dynamic Rule, and The Arrow's Alibi*
- *Khronon MOND and Khronon Cosmology*
- *One Field*
- *a_0 The MOND Scale*
- *Newton's G*
- *The Combination Rule*
- *BTFR*
- *The Cosmological Constant Λ*

§6 — The Unification Move: one arrow meets two demands

Sections 4 and 5 each ended on the same primitive. In §4 the arrow selects the quantum pointer basis; in §5 the arrow made dynamical — the khronon — is gravity's preferred time. This section makes the claim those two results were building toward: the basis-selector and the preferred-time are not two arrows that happen to resemble each other, but one primitive doing one job, seen in two sectors. That single identification is ED's answer to quantum gravity, and it dissolves the problem that has always kept the two theories from fitting together. It is the report's central claim, and it is argued rather than asserted: the section is explicit about how strong the argument is and where it stops.

The demand from each side

Quantum mechanics and general relativity make opposite demands about time, and the conflict has a name: the **problem of time**.

Quantum mechanics wants a preferred time. Unitary evolution runs in a time parameter; the measurement update happens *at* a time and distinguishes before from after; the definiteness of outcomes picks out a basis, and the theory is at its most natural when there is a fact about which time is flowing. General relativity forbids exactly this. Its diffeomorphism invariance says no foliation of spacetime is physically preferred; “which time is really flowing” is, in GR, a meaningless question. Canonical quantum gravity inherits the collision directly, the Wheeler–DeWitt equation famously has no time parameter at all. One theory needs the thing the other prohibits.

ED’s stance: side with quantum mechanics, and identify the time

ED does not split the difference. It sides with quantum mechanics: there *is* a fundamental preferred time. It is the arrow, the substrate’s direction of commitment (§3). And then it makes the move that turns a stance into a unification: **the preferred time quantum mechanics wanted is the same object gravity’s khronon foliation picks out**.

The two demands are met by one primitive. Quantum mechanics wanted a preferred time to make outcomes definite; §4 showed the arrow supplies it, selecting the channel basis. Gravity, in ED, is built on a preferred foliation; §5 showed that foliation *is* the arrow, dynamical. So the preferred frame of the gravity sector and the preferred basis of the quantum sector are not two independent structures that have to be reconciled. They are one structure, coarse-grained and divided into two sectors. The problem of time does not get solved by reconciling GR’s no-preferred-time with QM’s need for one; it dissolves, because ED is not a restating of GR, it has a real arrow, and that arrow is doing many jobs.

That is ED’s answer to quantum gravity, stated at the level this report can support: not a quantized metric, but a single primitive that is simultaneously the quantum pointer-basis selector and the gravitational preferred time.

Why this is more than relabeling

The obvious objection is that this is wordplay, calling two things “the arrow” and declaring victory. Three responses:

First, ED does not have two arrows to conflate. The reduction audit (§3, and the minimal-ontology paper it draws on) finds commitment to be the *sole process primitive*, the single source of anything dynamical in the theory. Under that finding, the measurement update and the gravitational evolution *cannot* be different arrows, because there is only one process. The identification is not a coincidence noticed after the fact; it is forced by there being one process.

Second, the single-arrow ontology actually works in both sectors. It is cheap to declare two things identical; it is not cheap for the identification to deliver, on one side, a functioning preferred-basis quantum theory (§4, einselection primitive, Born non-contextuality, the covering law) and, on the other, a gravity that reproduces the weak-field Einstein metric and unifies dark matter and dark energy (§5). One primitive carrying both working structures is the content.

Third, it dissolves a recognized tension rather than ignoring or papering one over. The problem of time is a real obstacle in mainstream quantum gravity, not a straw man. A framework that makes it *not arise* has done something, even before it has proven anything.

The black hole: where the conflict was sharpest

The place quantum mechanics and gravity are supposed to be most irreconcilable is the black hole, and specifically the information paradox: Hawking radiation looks purely thermal (it carries no information), evaporation looks complete (the matter is gone), and quantum mechanics is unitary (information cannot be destroyed). The

three statements cannot all hold. It is the sharpest test of any claim to unify the two, and the same arrow resolves it.

The resolution is the arrow doing §4's job one step further. Commitment is irreversible — that is the arrow — and **the arrow that forbids reversibility forbids erasure**: every commitment is a permanent, un-erasable record in the committed channel basis. Information is never destroyed at the substrate level, not because evolution is reversible, but because commitments are recorded and locked and the total bandwidth ledger is conserved (P04, bandwidth additivity). The infalling matter is committed and recorded as it participates near the horizon; those records straddle the horizon and survive through it. The horizon here is not new physics: it is the same $b \rightarrow 0$ decoupling surface — the boundary at which the controlled information capacity between substrate regions falls to zero, ED's A1 result (§12) — so the black hole is built from the report's own machinery, not a separate model. The paradox's load-bearing premise — that evaporation erases information — is simply false in ED, and no firewall is needed to protect a unitarity that was never the fundamental requirement.

On that picture the **Page-curve shape follows** — the radiation's entanglement entropy rises, turns over when the cross-chain entanglement budget saturates, and declines as the information re-routes into radiation-radiation and radiation-remnant correlations (Paper_050, Paper_052). Two caveats bound it. The turnover and the decline each rest on an *added* postulate — a finite V5 entanglement-bandwidth budget (for the saturation) and re-routing on saturation (for the decline) — which Paper_050 carries as postulates, not derived from the V5 kernel alone; so the shape is **derived-conditional** on that added structure, not a bare-substrate consequence, and Paper_050 is explicit that it does not derive the Page curve from first principles. That added postulate is, however, less ad hoc than it first reads: the finite V5 entanglement-bandwidth budget is now shown to be one and the same bounded envelope that sets the entanglement-monogamy bound and the quantum-error-correction plateau (§15), one budget seen in three arenas in fixed ratio, with the entanglement/boundary projection pinned to the Bekenstein–Hawking area law (a measured ≈ 0.88 cross-chain edges per Planck cell). The finiteness itself is form-forced by the kernel's boundedness; what stays inherited is the numerical scale (turnover near $t \approx 0.54 \tau_{\text{BH}}$). Two byproducts are testable: a sign-definite non-thermal $(\omega/\omega_c)^2$ correction to the Hawking spectrum, and a stable Planck-mass remnant where evaporation halts (§15).

The distinctive move: **ED preserves information without fundamental Hilbert-space unitarity** — which the arrow actually forbids. Reversible unitary quantum mechanics is the emergent, between-commitments description; the fundamental substrate is irreversible. The physically required property was never reversibility, it was no-information-loss, and irreversible recording delivers that directly. So the black hole is not an awkward case the unification must survive; it is where the arrow's defining character turns the paradox's central assumption on its head. (Tier: the resolution — records, no erasure, no firewall, emergent-not-fundamental unitarity — is grounded in bandwidth conservation (P04) plus commitment-irreversibility (P11), the substrate-level information ledger of Paper_052; the Page-curve shape and the Hawking and remnant predictions are derived conditional on the two V5 cross-chain postulates (the entanglement-bandwidth budget and its re-routing on saturation), which are structural additions; the interior reconstruction and the exact horizon geometry are the remaining computations. A standard Hilbert-space unitarity proof is not claimed, and in ED that is a feature, not a gap.)

This is the unification-bearing core of ED's black-hole sector, which is the part this overview carries. The fuller account — horizon thermodynamics, the Hawking spectrum and its $c_T = c$ -clean derivation, the Page-curve mechanism, the absence of a central singularity, and the Bekenstein–Hawking area law read as a count of straddling edges (the same holographic edge-count behind A1) — is the black-hole arc, roughly Papers 039–052, which this section points to rather than reproduces. The report is a map of the thirteen boxes a unified framework must check; the black hole is where the QM-and-gravity box is stress-tested, so it earns a place here, but its full treatment lives in the corpus.

The guardrail

This is a structural synthesis, an account, not a theorem, and its evidential force does not compound.

The minimal-ontology paper is explicit: the arrow's many jobs are “a pattern, not a compounding of evidence, seven conditional accounts do not sum to one hard result.” The same applies here at the two-sector scale. §4 is a reconstruction (with the Solèr residual); §5 is closed but leans on grounded, not derived, pieces in its dark

sectors. Identifying their arrows does not multiply their credibilities into something stronger than either. It exhibits a coherence: one primitive, worked independently in two sectors that mainstream physics treats as unrelated and even in tension, turns out to do a load-bearing job in each, consistently. Coherence-across-sectors is genuine evidence, it is the kind of evidence unification is actually made of, but it is weaker than a derivation, and the report says so.

Two specific limits. (i) There is no single theorem that *forces* “the einselection arrow equals the khronon arrow” from a deeper principle; the identification follows from the sole-process finding, which is itself a P11-*pivoted* audit result (the pivot is a disclosed input, not proven to be the only possible one). (ii) The identification currently carries no delivered prediction. If the pointer basis and the gravitational foliation are the same object, they should be locked together, and any observable consequence of that locking would be the substrate-scale signature §4 flagged as a candidate, currently unobservable (§15). Until such a signature is found and tested, this is a compelling structural picture, not a confirmed one.

A skeptic is right to press exactly here, and the right response is not to push back with more assertion. It is to point at the coherence and name its tier: **distinctive, structural, and unconfirmed**.

What this buys

§3 introduced the arrow; §4 and §5 described its two largest jobs; this section unified them. From here, the Standard Model sectors (§8–§11) are the same primitive again, in smaller and more inherited roles, the single hypercharge it anchors, the continuum it sorts, the spinor it undoubles, the chirality it half-carries. It is the primitive that keeps turning up as the mechanism, most strikingly where quantum mechanics and gravity were supposed to be irreconcilable. That recurrence, properly tiered, is the case for calling Event Density a unified framework.

Related papers. - *Commitment and Participation - Information-Paradox Synthesis, The Page Curve, Hawking Spectrum, and Horizon Decoupling*

§7 — Cosmology and the Dark Sector: the arrow at the largest scale

The arrow’s job here. §5 closed gravity as the khronon, the arrow made dynamical. At cosmic scale the same field writes the cosmology: it saturates into inflation, it bends galaxies as an interference with the horizon, and it leaves ED’s dark-matter question in a specific, honest place. This is the report’s frontier sector. One result here is native and strong, one is a genuine open with a named candidate, and one — the CMB — is the live edge where this whole class of theory is judged. It is stated at that mix of tiers, not smoothed into a win.

Inflation is the substrate saturating

Cosmic inflation, in ED, is not a new field on a tuned potential; it is the **saturation regime** of the substrate. In the post-boundary state the participation field Ψ is uniform ($\nabla\Psi = 0$), the kernel derivatives vanish, and the substrate stress-energy collapses to a vacuum-energy form ($w = -1$), which is de Sitter — the same exponential-expansion mechanism standard cosmology uses, reached from the substrate rather than an inflaton (Paper_ED_Cos_01; M3, form-identified / value-inherited). No inflaton particle exists at substrate level, and its detection would falsify the identification. This is **consilience, not a novel prediction**: ED reproduces the known sequence (the ~ 60 e-folds, the near-scale-invariant tilt) and *inherits* its values from Planck. The one genuinely native content is the ontological replacement — inflation as the substrate saturating, not a field slow-rolling — and the exit dynamics, how the uniform state breaks into structure, remain an open substrate-research item.

Dark matter is two questions, and ED answers “no particle” to only one

“Dark matter” is not one observation. Galactic rotation curves and the baryonic Tully–Fisher relation are one; the CMB acoustic peaks and cluster masses are another. They have resisted a single explanation for two decades — MOND fits galaxies and dies at the CMB, cold dark matter fits the CMB and is too clumpy in galaxies. ED’s

position is the split the data has been pointing at: **MOND where MOND works, a cold component where it does not.**

The distinction that matters, and that is easy to misread: ED's galactic result is that a dark-matter *particle* is not needed to explain galaxies. That is **not** the claim that no dark matter exists. ED's own gravity papers are explicit — *ED does not refute dark matter as particle content* (Paper_029, Paper_031). A particle is not needed *for galaxies*; the CMB is a separate question, and ED does not close the door on a component there.

MOND is an interference cross-term, not an added force

The galactic phenomenology of §5 has a cleaner reading than “a modification bolted onto gravity.” Reading gravitational strain as **quadratic in the participation amplitude** — $\text{Str} = |\sum_a P^a(a)|^2$ with $P = \sqrt{b} \cdot e^{i\pi}$ — splits it into a diagonal and an off-diagonal term. **The diagonal is Newton** (each source's self-potential); **the off-diagonal is MOND** — the interference *cross-term* between the local mass and the cosmic horizon, whose modulus is exactly the geometric mean $\sqrt{a_N \cdot a_0}$, with $a_0 = cH_0/(2\pi)$ supplied by the horizon (Paper_QuadraticStrain, sharpening Paper_030; Paper_029). This is the same interference structure as the double-slit, and it *discharges* the bilocal-coupling postulate the MOND line previously carried: the $\sqrt{}$ is forced, not chosen (Grounded / structural). So ED does not add a MOND force — MOND is what the interference of matter *with the horizon* looks like, and Newton is the same strain form's diagonal. One residual is open and named: the constructive sign of the interference, which is the same open question as the V5 kernel's attractive sign.

The CMB and clusters: a separate component, and the sector's live edge

MOND, being an interference *with the horizon*, handles galaxies with no dark component at all — but it does not supply what the CMB acoustic peaks demand: a source that gravitates without oscillating with the photon-baryon fluid, decoupled and diluting as a^{-3} , at $\sim 5\times$ the baryon abundance. This is where every MOND-class theory is judged, and KM-II is explicit that ED does not discharge it: no regulator is chosen, no CMB spectrum is computed, no cluster profile is fit.

ED's *candidate* for that component is native but unproven: a **committed neutral relic** produced at the inflation/saturation exit by the same admission mechanism that makes the baryons (Paper_ED_Baryogenesis), decoupled because it is uncharged, cold-capable because binding gives it mass (§11; Paper_MassWithoutMass), and diluting as a^{-3} . It shares the *mechanism* with MOND — the same interference cross-term, in two roles (matter-horizon is MOND, relic-relic is the dark component) — but it is **not** the same substance: MOND does galaxies without it, and the relic is a separate component for the CMB. Its one free number, the relic's free-streaming length (set by an *inherited* mass), must land warm enough to stay diffuse in galaxies, where a clumped relic would *double-count* against MOND — and the interference *amplifies* that double-count. That is the sector's live risk, stated plainly.

So the honest verdict: galactic MOND is native and now grounded as horizon-interference; the CMB dark component is an **open frontier with a native candidate and a named failure mode**, not a delivered fit. If the relic cannot be both warm-enough-for-galaxies and abundant-enough-for-the-CMB, or if a spectrum computed from the candidate misses the peaks, the cosmological half fails — and it could take the sector with it. The full, tiered map is physics-papers/dark-sector/.

What this buys

The arrow's largest-scale face is the same one it wears everywhere else: **inflation is the substrate saturating, and MOND is the arrow's horizon read as an interference**. What ED does *not* buy here is a solved dark sector; it buys an honest one — a native, elegant account of galaxies (interference, not an added force), an explicit refusal to over-claim (“no particle *for galaxies*,” not “no dark matter”), and a single sharp still-open question, the CMB component, carried at exactly that tier. This is the sector most likely to falsify ED, and the report says so.

Related papers.

- *Inflation as Substrate Saturation* (Cos_01)

- *Interference Gravity: Newton on the Diagonal, MOND on the Off-Diagonal* (Quadratic Strain)
- *Khronon MOND and Khronon Cosmology*
- *The Combination Rule, a_0 — the MOND Scale, BTFR*
- *Baryogenesis*
- *Mass Without Mass*
- The dark-sector position, debt map, and program: [physics-papers/dark-sector/](#)

§8 — Gauge Structure: the shape is derived, the multiplicities are the wall

The arrow’s job here. The arrow’s gauge-sector role is the single hypercharge. The Standard Model has exactly one gauged $U(1)$, and no conventional account says why one. In ED there is one because primitive P09 polarity is one primitive phased against one external flow, the commitment arrow (§3). Hypercharge is the arrow wearing a phase. This section describes the channel structure, and it sets a clean boundary: ED derives the *shape* of the gauge sector that the Standard Model postulates wholesale, and inherits the one piece the Standard Model also does not explain.

What the substrate fixes

Assemble the substrate into a bundle: the base is emergent spacetime, the fiber at each locus is the channel family (P07), the connection is polarity-transport (P05). The gauge group is the fiber’s structure group. For a family of N dynamically-indistinguishable channels carrying a complex amplitude $\psi \in \mathbb{C}^N$, the transport that mixes the channels while preserving total bandwidth (a P05 isometry) is the unitary group $U(N) = SU(N) \times U(1)$. So **non-abelian $SU(N)$ is the structure group of the channel fiber** — conditional, and the paper is precise about the condition: indistinguishability *alone* gives only the permutation group S_N ; the continuous $U(N)$ needs the $\mathbb{C}$ -amplitude (from the quantum-logic keystone, §4) and total bandwidth as the sole invariant.

On that structure the rest of the gauge sector’s shape follows:

- The **gauging is genuinely non-abelian**: position-dependent transport gives non-commuting holonomies, reconciled with einselection because the N channels within a family have no accessible label, so transport is free to mix them while the arrow pins the basis for the $SU(N)$ -invariant observables across families.
- The **action is F^2** : the substrate’s coherence-deficit on the plaquette holonomy is the Wilson action, $1 - (1/N)\text{Re Tr } U_p$, which coarse-grains to $-\frac{1}{2}\text{Tr}(F^2)$ — the Wilson trace being the matter-averaged per-chain deficit by Schur’s lemma, given the effective-action prescription. (The abelian case is grounded on the certified substrate; the non-abelian lift is analytic.)
- The **mass gap has a clean origin**: the same coherence-deficit carries the self-interaction $[A, A]$, which is zero for commuting (abelian) channels and nonzero for non-commuting ones. So in pure Yang–Mills, the classical massless flat direction is lifted exactly when the channels are non-abelian — Maxwell is massless because its channels commute, Yang–Mills gaps because they do not. This is the gap’s *mechanism*, verified at tree level; the continuum survival (asymptotic freedom, the Clay problem) is a different and harder question, not claimed.
- The **single hypercharge** is the arrow’s job above: one P09 phase against one arrow gives one $U(1)_Y$.

Put together: $\prod_i SU(N_i) \times U(1)_Y$. The Standard Model takes $SU(3) \times SU(2) \times U(1)$ as a postulate. ED derives that this is the *kind* of object the channel substrate produces — non-abelian, F^2 -acting, gap-carrying, with a single hypercharge — given its amplitude structure. That is a real accounting of structure the Standard Model simply assumes.

The wall: why {1, 2, 3}

What ED does not do is select the multiplicities. The framework gives $SU(N)$ for any N ; it does not say why the stable channel-families are singlets, doublets, and triplets and stop there. This is the report’s largest genuine open structural question (#1), and three things should be said about it plainly.

First, the channel-family stability does not select $\{1,2,3\}$ — a symmetric multiplet is stable for all N (the binding energy grows with N , the Hessian stays positive). ED does not currently derive the multiplicities.

Second, the Standard Model does not explain them either. The gauge group and its representation content are inputs there. So ED's status on this specific question — the shape derived, the multiplicities inherited — is not *worse* than the Standard Model's; it is the same inheritance, on top of a derivation of the surrounding structure the Standard Model does not attempt.

Third, the open question is bounded and well-posed, not a vague deficiency. It reduces to a single sharp question — why the internal channel-amplitude dimension is three — and it has one *candidate* route, still conjectured: the linking argument. Stable linking of one-dimensional loops exists in exactly three dimensions, and committed order needs a topologically stable holder, so ED plausibly reaches for 3D. But the corpus states two limits plainly. That argument targets the *spatial* dimension (P06). Whether that is even the *same* fact as the internal amplitude dimension of #1 is explicitly not claimed either way. So #1 is a well-posed open problem with a conjectured route, left unbuilt.

What is inherited or gated

- **Couplings and scales** (g , the effective graph scale a) set the normalization; inherited.
- **Charge values** (per-particle hypercharge) are the per-channel P09 advance rates; inherited. The single $U(1)_Y$ is grounded, but its charge normalization is not derived.
- **Electroweak breaking** ($SU(2) \times U(1)_Y \rightarrow U(1)_{EM}$, the Weinberg angle, the W/Z masses) rides on the substrate-Higgs and is Higgs-gated. The unbroken group is derived; the breaking is not.

Scope

- $SU(N)$ is derived-conditional on the $\mathbb{C}$ -amplitude, not from indistinguishability alone (which gives S_N).
- The F^2 action's non-abelian form is at the gauge-program (analytic) tier; only the abelian case is simulator-grounded.
- The mass gap is a mechanism, not a Yang–Mills existence proof; continuum survival is open.
- The single hypercharge is a grounded identification (P09 is one primitive), not a forced collapse.
- Spin- $SU(2)$ (the spacetime frame bundle) is a separate object, not this internal gauge structure (§11 handles the spinor).

What this buys

The rest of the Standard Model quarter will refer back to this section. The inherited casting of §10 (which force is chiral = $SU(2)$ pseudoreality), the inherited charge magnitudes of §9, and the inherited generation spectrum of §11 are all the *same* rep-spectrum wall, #1, and it lives here: ED derives the gauge sector's shape and inherits its multiplicities. Framed correctly, that is one bounded open problem, with a single conjectured route, sitting underneath a genuine derivation of structure — not three separate gaps, and not a vague failure.

Related papers.

- *The Gauge Structure of Event Density*

§9 — Charge and Electromagnetism: a native skeleton, a shadow field

The arrow's job here. Two of the arrow's roles from §3 show up together in the charge sector. Commitment irreversibility (P11) makes the substrate's phase winding single-valued, which is what quantizes charge and protects it. And the arrow's continuum-sorting job decides what electromagnetism *is* in ED: not a fundamental field, but the coarse-grained shadow the determinate substrate casts.

The skeleton is native and quantized

ED's charge result is reached graph-first: the participation graph is asked which topological invariants it carries, and only then is the survivor compared to charge. The survivor is the $U(1)$ holonomy on cycles. Committed single-valued polarity (P09 phase made single-valued by P11 commitment) forces its holonomy to an **integer winding** $w \in \mathbb{Z}$ — exactly $\pi_1(U(1)) = \mathbb{Z}$, to machine precision. That winding is **conserved and irreversibility-protected**: changing it would require uncommitting a polarity, which P11 forbids. And it sources a field topologically, obeying the **integral Gauss law** — the circulation around any enclosing loop is $2\pi w$, independent of the loop's size and shape. (This charge-as-topology result is Paper_ChargeAsTopology_B4, referred to below as **B4**.)

So charge quantization, conservation, and the unscreenable Gauss-law sourcing are structural facts of the ED graph, not inputs. Quantization is $\pi_1(U(1)) = \mathbb{Z}$; protection is the arrow. This is the topological skeleton of charge, and it is genuinely native.

The scope of this result is the skeleton, not the values. ED produces an integer winding with no selected magnitude — not the specific fractional charges (the Standard Model's fermion charges, quantized in thirds), not the fine-structure constant. Those live in the representation content, which the report inherits (§8, §14). §9's claim is about the *structure* of charge: that it is topological, quantized, and protected. The magnitudes are a separate question, and ED does not set them.

The smooth field is the coherence-weighted limit — and a shadow

Charge in ED established the skeleton but left one question open: the determinate substrate carries the integer winding and the Gauss law, but it does not produce a determined local $1/r^2$ field. The per-edge configuration is gauge/sweep-dependent; a determined isotropic Coulomb field appears only if one removes *both* orientation-blindness and P11 — at which point the system is ordinary lattice-field relaxation and no longer ED. The open question was whether the *coarse-grained* limit recovers Maxwell.

It does. The Maxwell action is already latent in the coherence term: $\cos^2(\Delta\phi/2) \approx 1 - \frac{1}{2}(\nabla\phi)^2$, so the incoherence is the standard $\frac{1}{2}(\nabla\phi)^2$ electrostatic action. Minimizing that coherence action around a point charge gives the **Coulomb field** — the FFT Poisson solve fits $\phi \sim A/r$ with $p = 1$ (3D Coulomb) at $R^2 = 0.97$, best among the tested exponents, with $\phi \cdot r \approx \text{const}$ in the near field (Paper_MaxwellEmergentShadow). So the coherence-weighted continuum limit of charge holonomies is Maxwell/Coulomb.

But that is a statement about the coherence-weighted ensemble, not about the determinate substrate. The certified substrate is kinetic and committal — it traps and commits, it does not relax toward a field-minimizing configuration (Paper_Continuum). It is Σ -blind to the phase sector. So the determinate substrate does **not** cast the smooth field: B4's open-edge analysis (its §8) resolves negatively for it. The smooth Maxwell field is the thick-limit **shadow** — form native (the charge skeleton is real substrate structure), smooth field coarse-grained (a came-back-no for determinate-dynamics, the same verdict ED reached for the diffusion PDE).

ED's monist position on electromagnetism

Put the two halves together and the sector has a clean ontology. The charge *skeleton* — winding, quantization, Gauss law — is native substrate structure. The electromagnetic *field* — the smooth A_μ , $F_{\mu\nu}$, the Coulomb $1/r^2$ — is not a second fundamental thing sitting beside the substrate; it is what the substrate looks like coarse-grained — like the smooth metric (§5, emergent and kinematic) and the diffusion equation before it. The determined continuum is always the coarse-grained object, never a primitive. ED does not have a fundamental field to reconcile. It has one substrate, and the field is its shadow. That is the same monist move the whole program makes, and electromagnetism is its cleanest instance, because the lattice-to-continuum dictionary here (Wilson's lattice gauge theory) is already standard.

Scope

- The winding is a structural realization of charge's skeleton. It is not a claim that the winding *is* electric charge, and no electromagnetic *content* is derived.
- The charge magnitudes and the fine-structure constant are not produced (inherited; the spectrum question is §8/#1).
- The skeleton results are on minimal certified arenas (rings, small grids); multi-winding interactions and coupling to matter fronts are not covered.
- The shadow result is analytic (the coherence-action minimizer is Coulomb) plus the structural reading (the determinate substrate does not cast it). It is not a certified-substrate simulation of the field, because the phase sector is Σ -blind, so there is nothing to run there.

What this buys

The arrow in the charge sector is: irreversibility quantizing and protecting the winding, and the continuum-sorting that makes electromagnetism a shadow rather than a fundamental field. Charge is the sharpest instance of a reading the whole report shares — the determined continuum, whether the smooth metric (§5) or the diffusion PDE, is always the coarse-grained shadow, never a second primitive. The representation content that would fix the charge *magnitudes* is the open rep-spectrum question.

Related papers.

- *The Topological Skeleton of Charge / B4*
- *Maxwell as an Emergent Shadow*
- *The Continuum: a Kinetic Lattice-Gas*

§10 — Chirality: ED builds the stage, inherits the assignment

The arrow's job here. Chirality is where the arrow can only do half the job, and the point of this section is saying which half. The chirality operator γ^5 factors, in ED, into the arrow multiplied by a spatial orientation, so the arrow *is* one of its two ingredients. And the arrow's parity-cleanness — the substrate has no reflection in its toolkit — is what forces the central theorem: the clean substrate is vector, and any parity violation must be spontaneous. What the arrow does *not* do is pick which force is chiral. That is inherited.

Parity violation, the fact that the weak force couples to left-handed fermions and not right, is one of the starkest facts in the Standard Model, and it is written into the law by hand. The question for ED is whether the substrate produces it or inherits it. The answer is precise: ED builds the entire stage on which chirality lives and inherits the one assignment.

The operator is grounded

The Dirac chirality operator is $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Read its two factors in ED. γ^0 is the arrow: ED's timelike direction is the commitment order (P11, retarded transport). $\gamma^1\gamma^2\gamma^3$ is the oriented volume element, a handedness of the spatial 3-frame that flips sign under reflection. So γ^5 is the arrow times a spatial orientation, both factors native. This grounds an operator the earlier parity work had found missing, and it explains why that work found "no local screw": γ^5 is not a per-channel object at all, it is global, the one arrow multiplied by the one global orientation.

The operator is only as real as its orientation factor, and two facts make it real. The orientation is **spontaneous** (derived): ED's rules are parity-symmetric, no primitive is a reflection, and a parity-symmetric rule set cannot fix a handedness, so any orientation the substrate carries is chosen by symmetry breaking, like a ferromagnet picking a direction. And the emergent space is **orientable** (structural): building a non-orientable space requires a reflection somewhere in the frame bundle, and ED has no reflection, so all frame relations are proper rotations and a single orientation propagates across the causal patch. The same one fact, no reflection primitive, does both jobs: it makes the orientation spontaneous and makes the space able to carry it globally.

The theorem: the clean substrate is vector

Model a family of N channels as a commitment map on an emergent chain, $H(k) = e^{\{ik\}}A + e^{\{-ik\}}B$, with A the forward (arrow-directed) hop, B the backward hop, k the wavenumber. Net chirality is the **point-gap winding** of this map, the number of times $\det H(k)$ encircles a reference as k runs its period; nonzero is chiral, zero is vector.

Parity-cleanness fixes B . Parity is the spatial reflection $k \rightarrow -k$ together with the lane reflection S of the channel row, and parity-symmetry of H forces the backward hop to mirror the forward one, $B = S A S^{-1}$. Then

$$S H(k) S^{-1} = e^{\{ik\}}B + e^{\{-ik\}}A = H(-k),$$

so $\det H(k) = \det H(-k)$: the determinant is **even** in k . An even determinant retraces its own path over the period, encircling nothing, so the **winding is identically zero, for every N and every forward hop A** . This was checked directly for $N = 1$ through 6 over many random hops, zero in every case.

The parity-clean substrate carries no net chirality at any channel-count. It is vector for one channel, two, three, all of them.

The corollary is the strong positive: since the clean transport is vector, any parity violation must break the parity symmetry, which means it must be the spontaneous orientation choice. **If parity is violated in ED, its direction is necessarily spontaneous** — not a handedness written into the law, because the clean rules provably cannot carry one.

Capability is not selection

Having a defined γ^5 is not yet parity violation; the dynamics has to treat the two handednesses differently. That splits into capability (is there a genuine left and right to couple to?) and selection (does the dynamics use it?). The capability structure is clean and precise: a parity-clean coupling that references the orientation must live in the reflection-odd part of the gauge structure. For $N = 1$ (abelian) that sector is empty, so the coupling is **forced vector** — which is electromagnetism. For $N \geq 2$ it is nonempty, so a chiral coupling is **possible**. There is a suggestive arithmetic here: reflect a row of N lanes ($i \mapsto N+1-i$), and a self-mirror central lane survives only when N is *odd* (a single lane, or the middle of three), so odd N is symmetrically anchored and reads vector, while *even* N — lanes that merely swap, with no anchor — can carry a handedness. That much is exact, and it correctly flags the weak force ($N = 2$) as the chiral one. But it is an intuition pump, not a law, and this report is clear about it. derived. The geometry says whether there is a left and a right lane; it does not say which the universe drives in. That is the whole content of the wall — and the real reason $N = 2$ is the special case is representation-theoretic, not a parity of N , as the next subsection explains.

The casting is inherited

Which channel-count actually ends up chiral, once the symmetry is broken, is not a transport fact. The clean winding is $\emptyset$ (the theorem); under a natural breaking it is N , monotone in the channel-count. The Standard Model's pattern is **non-monotone**: vector, chiral, vector for one, two, three channels. It matches neither $\emptyset$ nor N . So the transport dynamics does not select it.

Where the selection genuinely lives is representation theory — and it explains why the lane arithmetic *hits* while its even/odd extrapolation *fails*. The fundamental representation of $SU(2)$ is **pseudoreal**: the doublet is equivalent to its own complex conjugate, $2 \cong \bar{2}$, tied together by the antisymmetric invariant $\epsilon_{\{ab\}}$ (the same object that makes $SU(2)$ the double cover $Sp(1)$, and that gives the spinor its metric). The fundamentals of $SU(N \geq 3)$ are **complex** — N is *inequivalent* to $\bar{N}$ — and this is what singles $N = 2$ out *uniquely*, not any parity of N : $N = 4$ is complex, like $N = 3$, not self-conjugate like $N = 2$. That self-conjugacy is the real content of “two channels is special,” and it is a standard fact of group theory (see Paper_CleanSubstrateVector; the lane test is foundations/T4_12). ED **inherits** it, exactly as it inherits the gauge multiplicities and the constants (the same rep-spectrum question as §8 and §14). And group theory here is *permissive*, not deciding: nothing forces $SU(3)$ to be vector — its complex 3 could have carried a chiral theory, and nature simply put both quark handednesses

in the 3 — so the vector/chiral casting is contingent matter content the Standard Model stipulates and ED takes as given. The contribution here is to name precisely what is behind the wall: $SU(2)$ pseudoreality.

The prediction, and the anomaly baseline

Because the clean substrate is provably vector, ED's parity violation is necessarily a spontaneous, per-universe orientation tied to the arrow, and the same arrow sets the matter/antimatter sign (§11). So ED predicts the **gauge handedness and the matter/antimatter sign are correlated** — two faces of one first-commitment choice — against the Standard Model's fixed, uncorrelated, law-level handedness. The claim is theoretically firm (it rests on the vector theorem, not a stance), but its **testability is open**: the corpus register flags it as not testable with current data, so §15 carries it as a distinctive-but-not-yet-testable prediction rather than a live weapon.

The theorem also fixes the anomaly baseline: a vector theory is automatically anomaly-free, so the clean substrate carries no gauge anomaly (at the transport level, contingent on the substrate-to-Dirac descent). The Standard Model's nontrivial chiral cancellation is a property of the inherited chiral content, so it is inherited with that content (§13).

Scope

- The casting is inherited, not derived. The wall is explained, but not breached.
- The theorem is at the channel-transport level; the relativistic descent to the full Dirac sector is the standing open arc (§11, §14).
- No representation spectrum, hypercharges, masses, or charge magnitudes.

What this buys

The arrow does half the job of chirality cleanly: it is half of γ^5 , and its parity-cleanness forces the vector theorem, so parity violation is provably spontaneous rather than law-level. The other half — which force is chiral — is the representation-theoretic inheritance behind the rep-spectrum wall. And the correlated-handedness prediction is a distinctive claim (testability open, §15). The verdict is a clean division: ED builds the stage of chirality and inherits the one assignment.

Related papers. *The Clean Substrate Is Vector* which superseded *The Parity Wall*

§11 — Matter: the spinor, and mass without mass

The arrow's job here. The matter sector uses the arrow four ways. The arrow **undoubles** the lattice spinor, cutting the Nielsen–Ninomiya sixteen species to one. Its first commitment **selects matter over antimatter** (the charge-conjugation reference C), the one chirality-adjacent thing ED fixes natively. The arrow makes a lone excitation **massless**, moving always at the substrate speed, which is what forces mass to come from binding. And the arrow's commitment rate sets a **clock**, which turns out to be time dilation and not mass, which this section separates.

The matter sector splits into two questions: does ED produce the Dirac spinor, and does it produce mass? The spinor is form-complete and structure-inherited, the standard ED shape. Mass is the more interesting result, because ED accounts for it exactly where most of it actually lives.

The spinor: substrate to Dirac, form-complete

The Dirac operator's *form* is forced: it is the unique first-order Clifford-linear factorization of the substrate's Klein–Gordon operator, $D = i\gamma^\mu \partial_\mu - mc/\hbar$. Its *substrate continuum limit* is now computed, not asserted: retarded transport on the participation graph gives $D(p) = \sum \gamma^\mu (e^{ip_\mu} - 1) \approx i\gamma^\mu p_\mu$ near $p = 0$ (Paper_106). And the **arrow undoubles it**: a naive Hermitian lattice Dirac operator has sixteen doublers, and the arrow's retardation is exactly the Wilson term that removes them (paying its standard price, explicit

chiral-symmetry breaking, here carried by the arrow's non-Hermiticity), leaving a single species at the origin, verified directly (`chiral_3p1d.py`). This is ED's concrete escape from Nielsen–Ninomiya, a relational graph plus a genuine (non-Hermitian) arrow.

The undoubling script (`chiral_3p1d.py`) settles the *count* ($16 \rightarrow 1$), not the handedness; it explicitly flags that whether the survivor is chiral or vector-like is an open relativistic question it does not resolve. That was addressed separately by the substrate-to-Dirac analysis (T4_04), which found the survivor **vector-like by default** (a *derived-conditional* verdict, not a proof: the channel topology could still be handed). That is the right outcome, because a Dirac fermion *is* $L \oplus R$ and *which* handedness the weak force uses is the inherited casting of §10. So the vector-like survivor is a feature of a correct Dirac reduction, not a gap, but the verdict comes from the T4 analysis, not from the doubler count.

What is genuinely inherited is narrower than it looks: the metric *signature* (Lorentzian 3+1, via the acoustic-metric continuum limit) and the mass value. Given that signature, $C(3,1)$ is **not** a further inheritance — it is the spinor-supporting Clifford algebra **fixed by that signature** (form-identified, Paper_103, RQM-T2), and the Dirac operator is its Clifford-linear factorization: Clifford is the only structure that linearizes the second-order wave operator, which was Dirac's original point. What remains standard mathematics is only the internal representation-theory machinery (the unique four-component irreducible representation, via Pauli's theorem and Schur's lemma). The one route that would make the spinor fully substrate-native — building the 4-spinor from graph degrees of freedom rather than the imported algebra — runs into the same channel-topology wall as the representation spectrum (§8, §14): canonical channels carry no topology to be a spinor's, and the program that would supply one is unbuilt. So the spinor is form-complete, with only its signature and mass inherited and its algebraic structure form-identified, its one deepening blocked at the rep-spectrum wall.

Matter over antimatter

The spinor is vector ($L \oplus R$), so on its own it does not distinguish matter from antimatter. The arrow does. Charge conjugation C is a reference that has to be set somewhere, and in ED it is set by the first commitment: first arrival selects one side, and matter is here rather than antimatter because matter committed first. There was no annihilation event. This is the one chirality-adjacent thing ED fixes *natively* (parity, the handedness, is inherited — §10), and it is the same object that appears in §10's operator grounding, where $\gamma^5 = \text{arrow} \times \text{orientation}$ has the arrow as its C -side ingredient and the orientation as its P -side. So C and P are two attributes of one operator, both set at the arrow's early action, and C is the half ED derives. It carries no matter/antimatter *number* (no baryon asymmetry magnitude); it is the reference, native, that the number would be measured against.

Mass looks impossible for a single front

ED's certified rule is **ballistic-or-extinct**: a front advances one hop at the maximal speed or it dies. There is no dwell, no fractional hop, no slow-but-surviving mode. Rest mass needs exactly that missing mode, something that lets a thing sit slower than c . So a lone ED front cannot have rest mass; it moves always at c .

That looks like a failure, but a reframing turns it into the mechanism: a lone front shares the cosmic horizon's defining feature, c -moving, no rest frame, massless. Mass is therefore not a property a front can carry. If mass exists in ED, it has to come from what a *collection* of fronts can do that one cannot, bind. And this is how most real mass already works: roughly ninety-nine percent of a proton's mass is binding energy, not the Higgs coupling; a box of light has rest mass though every photon in it moves at c .

Mass from binding, measured

The test is whether ED's cross-chain coupling V5, in its known finite-reach retarded attractive form, confines massless fronts into a bound composite. On the certified simulator it does (`Paper_MassWithoutMass`):

- A **free front** moves at c ($v = 0.98$).
- With the coupling **off**, a cluster disperses (extent $28 \rightarrow 55$): unbound.

- With the coupling **on**, the cluster is **confined** (extent stays 1.4 – 2.3) while each constituent keeps moving at c , and the composite's center of mass is **sub-luminal** ($v \approx 0.5 < c$). This is a genuine bound state of massless-moving parts.

The composite has the defining property of mass, **inertia**. Under a uniform force the bound composite responds at $v_x = 0.72$, while an *unbound* cluster of the same size responds at 0.97 ($\approx$ the free front). The resistance is from binding, not from averaging, and that gap is the controlled, load-bearing result. And the composite heads toward rest as it grows: the center-of-mass drift shrinks monotonically with constituent count (0.54 $\rightarrow$ 0.31 from eight to thirty-two), the internal momenta cancelling better in a larger bound state.

This is **mass without mass**: a bound system of c -moving constituents has rest energy, hence rest mass, and it is the physically dominant form of real mass. It is native to ED's dynamics, conditional on V5, which is a faithful structural addition (finite-reach, retarded, attractive, matching the corpus kernel) but not shown to be forced by the bare primitives.

Mass is not time dilation

There is a *second* way to make something move slower than c , and separating it from mass is what makes the mass result clean. It is **not** the certified rule — that rule is ballistic-or-extinct, with no dwell (above). It is a deliberately-added, *non-certified* mechanism, brought in only as a foil: a front *given* commitment-memory **dwells**, re-committing in place, which lowers its advance rate. Does that slowdown amount to mass? No. Push such a memory front and its forward drift tracks its path speed exactly ($v_x/\text{path} = 1$ at every memory level): there is **no directional inertia**. A slow clock, not a mass. This is **time dilation** — the same commitment-rate factor that appears in gravity's sparse-commitment parameter (§5), which is why gravitational time dilation shares it. Only *binding* produces the directional inertia ($v_x/\text{path} < 1$) that is mass. So the two statements that looked contradictory are consistent: the certified fronts never dwell, the mass comes from binding those dwell-free fronts, and the memory-dwell — a deliberately off-certified foil — exists only to show that a mere slow clock is time dilation, not mass.

So two things the substrate had entangled come apart cleanly: **commitment sparseness sets the clock (time dilation, and it ties matter to gravity through the shared factor); binding sets the mass**. They are distinct phenomena, and the arrow is behind both, once as the commitment rate, once as what makes the bound constituents massless in the first place.

Scope

- Binding mass is native but **V5-conditional**: V5 is a structural addition, not shown primitive-forced (the bare substrate's native cross-chain coupling was separately measured to be dispersive).
- This is **binding** mass, the dominant form. The **fundamental Higgs/electroweak mass** (electron, current-quark, W/Z from spontaneous breaking) is a separate mechanism, inherited; the condensate route to it comes up empty on the certified field.
- The rest limit is an extrapolated size-trend, not a measured zero; the equivalence-principle reading of the uniform-force response is a consistent interpretation, not a proof.
- No numerical mass values, and no generations, masses, or mixings, those are the inherited value layer (§12) and the rep-spectrum content (§8, §14).

What this buys

The matter sector is the arrow four times over: undoubling the spinor, matter over antimatter, making the lone front massless (so mass must be binding), and setting the clock that is time dilation. The mass-from-binding result is a genuine native mechanism for the dominant form of real mass, measured on the certified substrate, and the clean mass/time-dilation split reuses the same commitment factor that runs gravity's clock, tying the matter and gravity sectors together through the arrow once more. The pieces that stay inherited — the metric signature, the fundamental Higgs mass, the generation spectrum — all point to the same two places the report flags: the value layer and the rep-spectrum wall. (The Clifford algebra itself is fixed by the signature — form-identified, not inherited; Paper_103.)

Related papers.

- *The Dirac Equation*
- *Cl(3,1) Frame Uniqueness*
- *Mass Without Mass*

§12 — The Constants: inherited by ED’s own logic

The Standard Model has roughly twenty free parameters — the fine-structure constant, the fermion mass ratios, the mixing angles, the cosmological constant’s value (the scale H_0), the gauge couplings — and it derives none of them. ED does not derive them either. This section argues that this is not a hole but rather a structural feature of any relational ontology.

The structural reason

ED’s determinability boundary (A1) established a sharp fact about the substrate: the only observer-independent scalar the substrate produces in isolation is **zero**. Controlled channel capacity between regions is exactly zero (ED’s A1 theorem, derived, not posited). A substrate whose ontology is pure relation has no place to store an intrinsic dimensionless number. The corpus’s own image for this is exact: asking the substrate for the fine-structure constant is like asking one molecule for the temperature of a gas. Temperature is real, but it is a property of an ensemble, not of a fragment, and no amount of interrogating the fragment yields it.

So the constants are not quantities ED is failing to compute. They are quantities of a kind the substrate provably does not carry in isolation. The values are global, relational facts, inherited from measurement, and the framework’s own structure is what says they must be.

The taxonomy of what ED can and cannot produce

This is not a convenient after-the-fact story; it matches exactly what ED’s machinery does and does not do. Every structural move in the corpus produces an **O(1) coefficient sitting on top of an already-inherited scale** — the holographic area count gives the $1/4$ in $S = A/4$ on top of ℓ_P ; the near-horizon smoothness gives the 2π in the Hawking temperature on top of κ ; the mixing-matrix phase count gives the combinatorial $(n-1)(n-2)/2$ with the numerical inputs still inherited. Nothing in the demonstrated toolkit produces or suppresses across orders of magnitude.

And where ED tried to derive a genuine dimensionless number, it failed. The fine-structure constant was attempted by three independent routes (topological winding, an RG fixed point, a cross-overlap integral) and all three failed. The fermion mass ratios were attempted by six mechanisms and all six were refuted (a single real scalar field carries no topological invariant to quantize them). These negatives are exactly what the structural claim predicts: ED produces forms and O(1) factors, not the values.

The cosmological constant: a dissolved catastrophe, not a failed derivation

The cosmological constant looks like it should be the worst case — the notorious 122-orders-of-magnitude gap between the Planck-scale vacuum energy quantum field theory predicts and the tiny value observed, “the worst prediction in physics.” What made it catastrophic was the *infinite tower of modes*: summing zero-point energies up to the Planck scale gives $\sim M_P^4$. ED’s substrate is discrete, so there is no infinite tower to sum, and that $\sim M_P^4$ vacuum energy is structurally absent (Paper_038_5). The huge number was the size of the mistake, not a real quantity waiting to be cancelled — so the *naïve* catastrophe does not arise. (Paper_038_5 is careful that this is a *partial* dissolution: a residual hierarchy survives, taken up below.)

What survives is a genuine ED contribution, carefully bounded. *Λ’s form* is **identified** — the structure is fixed (a vacuum-energy density set by the universe’s reach and the substrate’s grain, the V1 kernel integrated over the cosmic horizon $R_H = c/H_0$), though as a composition of inherited pieces rather than a from-nothing derivation. And the smallness is *reframed*, not fine-tuned: it is the dimensionless Friedmann ratio $(H_0/M_P)^2 \approx 10^{-122}$,

what two ordinary scales give, rather than a Planck-density vacuum cancelled to sixty decimal places. What is **not** claimed is a derivation of Λ 's *value*: it reduces to the one inherited scale H_0 (with M_P substrate-side via Paper_027), and pinning H_0 from the substrate — closing the residual hierarchy — is the open Route-A derivation, which Paper_038_5 states plainly is open. So Λ is *value-inherited via H_0* , like every other constant, while its notorious *catastrophe* is dissolved and its form identified.

The resting place

The consistent conclusion, reached in the corpus and adopted here, has two parts that must be kept distinct. First, c , ℓ_P , and $\hbar$ are **dimensionful units**, not numbers ED is failing to compute: they are conversion factors one can set to 1 — in ED, c is one grain per tick, ℓ_P is the grain (P08), and $\hbar$ is the action quantum — and only *dimensionless* combinations of constants carry observer-independent physical content at all. Second, that physically-meaningful content, the genuinely inherited layer, is the **dimensionless** set: α , the mass ratios, and the cosmological ratio $\Theta_{ED} \approx (H_0/M_P)^2 \approx 10^{-122}$. It is *that* kind of number the substrate provably cannot carry (the A1 result). Declining to chase it is a taxonomic verdict — no ED move produces a dimensionless number — not an instinct.

Two clarifications bound the claim. First, dissolving the *naïve* mode-tower catastrophe and identifying Λ 's *form* (above) is not the same as deriving its *value*: that still reduces to H_0 , an inherited substrate scale whose first-principles derivation (Route A) is open, and the residual hierarchy closes only with it. ED inherits the number; what it removes is the infinite-tower catastrophe and the mystery of the form, not the last measured input. Second, the “no intrinsic scalar” claim is a structural posture with strong support (the A1 zero-scalar result, which is derived, plus the taxonomy of successes and failures), not a from-nothing impossibility theorem. It is as firm as A1 and the failure record make it, and it is stated at that tier.

The fair comparison

The decisive point for a skeptical reader is the same one that governs §8. The Standard Model does not derive its constants; they are its free parameters, fit to data. ED incorporates an identical inheritance. What ED adds on top is the structure the Standard Model leaves unexplained: the *form* of Newton's constant ($G = c^3 \ell_P^2 / \hbar$), the *form* of the MOND scale ($a_0 = c H_0 / 2\pi$), the *form* of the cosmological constant ($\rho_\Lambda = (3/8\pi) H_0^2 M_P^2$, §5), the combinatorial *form* of the mixing-matrix phase count, the F^2 action, the single hypercharge. ED organizes the constants and derives the relations among the forms; it inherits the numbers, exactly as its ontology says it must, and exactly as the Standard Model does anyway.

What this buys

The constants are the report's — column, and they belong in a different category from the two structural opens. #1 (the rep-spectrum) and #2 (the chiral anomaly cancellation) are things ED *could* still derive and has not; the constants are things ED's own logic says it *cannot* and *need not* derive. Keeping that line sharp is what stops the twenty free parameters from reading as twenty holes. They are one principled inheritance, predicted by the substrate's lack of an intrinsic scalar, and no worse than the Standard Model's own standing on the same numbers.

Related papers. *Common Cause, Not Channel / A1 The Cosmological Constant Λ*

§13 — Internal Consistency: the anomaly question, split by tier

The arrow's job here. Two of the arrow's results feed this section. Its parity-cleanness makes the clean substrate vector (§10), which fixes an anomaly-free baseline. And the anomaly-relevant non-Hermiticity, the thing a genuine chiral anomaly would live in, is the worldline arrow itself (a point-gap spectral flow). So the arrow both sets the safe baseline and is where the one open question lives.

Gauge anomaly cancellation is the deepest internal-consistency requirement of a chiral gauge theory: the gauge current must stay conserved at the quantum level, and in the Standard Model this works only because of a strik-

ing numerical balance among the hypercharges. The question is what ED contributes to this, and the answer is not “ED forces it”. The answer is that the question splits, exactly the way the chirality casting did, into a solid side and an inherited side, with one narrow open candidate.

The solid side

Two pieces are secure.

Charge conservation. The charge-as-topology result (§9) gives an exact integral Gauss law, circulation = $2\pi w$, loop-independent, to machine precision. An exact Gauss law is exact gauge-charge conservation, which is the substrate statement of anomaly-freedom for the charge sector. (Read with charge’s own caveat: the winding is Σ -blind and weakly coupled, so this is analytic, and reading the exact Gauss law as anomaly-freedom is an interpretive step on top of the measured conservation, taken here.)

The clean baseline. A gauge anomaly requires chiral content — left and right contribute with opposite sign, and a vector-like theory cancels automatically. §10’s theorem is that the parity-clean substrate is vector for every channel-count. So the clean substrate carries no chiral spectral flow, and there is nothing there to threaten the exact Gauss law. **The clean baseline is anomaly-free, trivially and provably** (at the transport level, contingent on the substrate-to-Dirac descent, like the theorem it rests on).

The inherited side

The Standard Model’s anomaly cancellation is *nontrivial*: the theory is chiral, and its consistency depends on the special balance $\Sigma Y = 0$ and $\Sigma Y^3 = 0$ per generation, the condition that ties the quarks (times three colors) to the leptons. This is one of the Standard Model’s most striking just-so facts, and a grand unified theory like $SO(10)$ explains it by putting a generation in a single **16**.

In ED, the chiral content appears only after the spontaneous symmetry breaking of §10, and *which* chiral content it is, is the inherited casting (the pseudoreality of $SU(2)$). So the nontrivial cancellation is **inherited along with the content**: ED takes the chiral matter content as given, and its anomaly-freedom comes packaged with it. Calling this “inherited” does not contradict the report listing the rep-spectrum as *open* (#1) — the two are the same content viewed two ways. Inherited is where it stands today; #1 (§8, §14) is the derivation that, if it closed, would produce that content and its anomaly-freedom together. So the statement is narrow: ED does not *today* independently derive why the emergent chiral content is anomaly-free. That would follow either from closing #1 (deriving the content), or from the separate, non-inherited candidate below (#2).

The one open candidate

ED’s characteristic move is to force a *constraint*, not a *value*, and there is exactly one place it might do so here. ED’s Gauss law is *exact*. If that exact charge conservation must survive the arrow’s chiral spectral flow, then the substrate would **forbid** a gauge anomaly by construction — forcing *that* whatever chiral content emerges is anomaly-free, without ever selecting *which* content. That would be a genuine, non-inherited ED contribution to the deepest consistency condition in particle physics.

It is a candidate, not a result, and it is gated. Establishing it requires the substrate-to-Dirac worldline reduction (T4, §11), because a gauge anomaly is properly a relativistic object and the spectral-flow face lives there. The operator triage is complete — the anomaly-relevant non-Hermiticity is specifically the worldline arrow, and every off-worldline candidate (the V1 kernel, V5, the Lindblad effective Hamiltonian) was checked and ruled out for a distinct reason — so the question is correctly located, but the reduction that would answer it is the one deep arc still open in the matter sector. The candidate is named, not claimed.

Scope

- The conservation face and the clean baseline are solid; the nontrivial chiral cancellation is inherited; the constraint-forcing possibility is a T4-gated candidate.

- All of it is at the transport/structural level; the relativistic anomaly proper is gated on the substrate-to-Dirac reduction.
- ED does not derive the Standard Model’s anomaly cancellation. The section maps the question to its floor.

What this buys the report

The anomaly box is the report’s second genuine structural open (#2), and it is smaller and better-located than #1. Its solid pieces are real (exact conservation, a provably vector baseline), its inherited piece is the same rep-spectrum wall the rest of the Standard Model quarter inherits, and its one non-inherited possibility is precisely gated on the T4 reduction that also gates §10’s relativistic chirality and §11’s graph-native spinor. So #2’s anomaly constraint-face is gated on the substrate-to-Dirac descent — the same reduction that gates §11’s spinor, though on a different residual of it (the anomaly notes route it to the non-Hermitian→unitary / covariant-worldline piece, not the channel-topology residual that walls #1) — and it is left open.

Related papers.

- *The Topological Skeleton of Charge / B4*
- *The Clean Substrate Is Vector*

§14 — The Walls: what ED does not reach, and of how many kinds

A unified framework is judged as much by the sharpness of its stated limits as by its results. ED’s limits are not one undifferentiated “not yet.” They come in distinct kinds. There are four: one proven wall, two bounded structural opens, a set of principled inheritances, and (separately, §12) the constants. This section states the first three plainly.

The one proven wall: primality

ED has exactly one limit that is a *theorem*, and it is worth stating first because it is the strongest evidence that ED’s walls are real and not rhetorical.

ED’s substrate is, by its primitives, in the finite-reach / finite-memory class: every determination is a function of a bounded local residue state, which is a periodic, zero-entropy sequence. Using the integers as an external ruler, the prime structure splits into a **template** (the finite-local residue statistics — coverage by small primes, equidistribution among open residues) and an **escape** (primality itself, the Möbius sign, pair correlations). The finite-memory class reproduces the template exactly — the scale-invariant 1.700-bit template invariant and the least-prime-factor density ladder are matched to five decimals — and **provably cannot reach the escape**. The optimal finite-memory correlator with the Möbius function decays to zero for every fixed memory ($\sim \sqrt{M/N}$), the per-open-position primality entropy is irreducible, and the twin density requires a pair-correlation the template does not supply. The mechanism is a $\sqrt{N}$ activation horizon: the memory needed to resolve more grows with the scale it must resolve, so no fixed finite memory ever catches up.

What makes this a *proven* wall rather than a located one is that it is anchored to established theorems outside ED — Möbius orthogonality to periodic sequences (the prime-number theorem in arithmetic progressions) and the sieve parity barrier — not to ED grading its own homework. ED belongs to the class those theorems constrain, so the ceiling is located by theorem.

Two things frame this wall. It is not a failure: ED is not in the business of generating the primes, and the integers are a ruler, not a target. And it is a strength, not a weakness — a framework that can name exactly something it provably cannot do, and cite the theorem, is demonstrating that its other walls are drawn with the same seriousness. This wall is also the external twin of A1 (§12): the same finite-reach ceiling, measured once internally (controlled channel capacity is exactly zero) and once against a math-certified ruler.

The two structural opens (located, not proven)

Two limits are genuine open problems — things ED *could* still derive or delegate, not things proven impossible. These are the report's real **!s**.

#1, the representation spectrum. Why the stable channel-families are $\{1,2,3\}$, and behind it, why the internal amplitude dimension is three. ED's natural route (channel-family stability) was tested and refuted, and that is reported as a failure (§8). What remains is bounded but *conjectured*: the question reduces to a single sharp one (why internal-d = 3), whose one candidate route is the linking argument — stable linking of 1D loops is unique to 3D, and committed order needs a stable holder. Two caveats: that argument targets the *spatial* dimension (P06), and whether it is the same fact as the internal dimension is not claimed; and its open premise (that committed order is held by linking *specifically*) is conceptual and not even askable on the certified 2D simulator. So it is a candidate route, not a delegatable computation. And it is unsolved in the Standard Model too, which postulates the multiplicities.

#2, the chiral anomaly cancellation. Whether ED's exact Gauss law forces anomaly-freedom on the emergent chiral content, ED's one possible non-inherited contribution to the deepest consistency condition in the Standard Model (§13). It is gated on the substrate-to-Dirac worldline reduction (T4). But that reduction is itself now *form-complete*: the Dirac operator's form is forced, its substrate continuum limit is computed, and the arrow's undoubling of the lattice spinor is verified (§11). Its *graph-native-spinor* residual — building the 4-spinor from the participation graph's own channel degrees of freedom — is blocked at the same channel-topology wall as #1. But #2's anomaly face does *not* sit on that residual: the anomaly notes route it to a *sibling* residual of the same reduction, the non-Hermitian→unitary / covariant-worldline piece, which is not shown to be the channel-topology wall. So the two structural opens are two residuals of *one reduction* (the substrate-to-Dirac descent), not two separate arcs and not one wall: #1 reduces to the channel-topology wall, #2 to a sibling residual, plus the rep-spectrum's conjectured linking route. Two well-located problems sharing a root reduction, not a diffuse fog — but genuinely two residuals, not one.

The principled inheritances (settled, not open)

Two further limits are neither proven walls nor open problems: they are resolved as inherited, with reasons.

Dimension. Three-dimensionality (P06) is selected, not derived. But it is selected independently three ways — the internal amplitude dimension from the gauge content, the holographic reach law from observed gravity, and a linking argument from the substrate's own topology — and the linking route is a genuine derivation *candidate* that would route 3D back through the arrow (committed order held stable by linking, which is unique to three dimensions). So dimension is inherited with a conditional route home, not a blank.

Casting. Which force is chiral is resolved-as-inherited: the pseudoreality of $SU(2)$'s fundamental versus the complex representations of $SU(N \geq 3)$, a representation-theoretic fact ED carries (§10). It is not an open ED derivation and not a wall to break; it is the same rep-spectrum inheritance as #1, correctly located outside ED's dynamical output.

The map of “no”

ED's limits sort cleanly:

- **Proven wall (1):** primality — theorem-anchored, the finite-memory ceiling. ED knows at least one thing it provably cannot do.
- **Structural opens (2):** the rep-spectrum (#1, a conjectured linking route to the channel-topology wall) and the anomaly constraint (#2, gated on a *sibling* residual of the same substrate-to-Dirac reduction, not the same wall as #1). Both bounded, both unsolved in the Standard Model, both residuals of one reduction that is itself form-complete (§11) — one root, two distinct residuals, not a single wall.
- **Principled inheritances:** dimension (selected, conditional route home) and casting (rep-theoretic, resolved-as-inherited).
- **Inherited-by-design (§12):** the constants — a fourth category, things ED's own logic says it cannot and need not derive.

What this buys

The walls set up the two sections that close the report. §15 asks the harder question — not what ED has not yet derived, but what could prove it *wrong* — and §16 states the bottom line. The map here is what lets those sections be read as confident rather than defensive: a framework that has located its one proven limit, bounded its two real opens, and named its inheritances has earned the right to state plainly what it does and does not claim.

Related papers.

- *Template, Not Escape*
- *Common Cause, Not Channel* / A1 *
- *The Gauge Structure of Event Density*

§15 — The Falsifier Frontier: what could prove ED wrong

A framework earns the word “unified” by explaining, and it earns the word “scientific” by being killable. This section asks where ED diverges sharply enough from Λ CDM, standard quantum mechanics, and general relativity to be proven wrong at a measurable point, and it states the status of the one thing that would end the argument: **no distinctive, argument-ending prediction is confirmed yet.**

What counts as a weapon

The corpus draws a line that this report keeps. A **weapon** is a prediction that is *novel/distinctive* (standard physics does not make the claim or measurably diverges from ED), *sharp* (a number, a sign, or a shape, not a direction), and *near-term testable*. A **postdiction** — reproducing an already-known value — is consilience: it supports the framework but cannot end an argument, and several such postdictions are shared with Λ CDM, MOND, or GR. An internal-consistency check is not an empirical discriminator at all. Because ED inherits the value layer, no weapon here is a predicted *number* from first principles; the weapon is always the *form, sign, or shape*, as that is what can fail.

The ranked weapons

These are ED’s sharpest falsifiable bets. The sharpest *live* one, and the only weapon here with distinctive data already in hand, is the redshift evolution of the MOND scale, stated first below. The rest follow in the corpus’s own priority order (which predates the f_v correction and the cluster-knee revision below, so read the numbering as the inventory’s, not a fresh re-ranking).

★ **The flagship: the redshift evolution of the MOND scale, $a_0(z) = cH(z)/(2\pi)$.** Because ED ties a_0 to the cosmic horizon $R_H = c/H$, the MOND acceleration scale is not a constant of nature (as in MOND) but tracks the instantaneous Hubble rate, so it must grow with redshift, roughly $\times 1.8$ by $z = 1$ and $\times 3$ by $z = 2$. The power $\alpha = 1$ is forced — though, to be precise, not by dimensional analysis: ED carries a second length scale (ℓ_P), so $a_0 = c \cdot H \cdot (\ell_P H/c)^p$ is dimensionally legal for any p . What fixes $p = 0$ is the projection mechanism (Paper_029’s dipole projection introduces no ℓ_P) together with ED’s no-intrinsic-scalar result (§12): with the $\sim 10\%$ magnitude match already in hand, ED has no order-spanning dimensionless number available to bend the power. So $\alpha = 1$ is mechanism-forced, and ED cannot fit a_0 the way MOND fits its constant. The prediction is genuinely forward-dated — it is in Paper_038’s cosmological section two months before the data, with non-evolution named as a refutation — with one caveat of record: a second corpus paper (Paper_031) disclaimed the evolution prediction until that internal contradiction was reconciled on the day the data was confronted. And the first data exist. MUSE-DARK III (A&A 2026) measures the radial-acceleration-relation scale across $0.33 < z < 1.44$ and finds it evolves at $\sim 30\sigma$, with a_0 rising from 1.2 locally to $2.38 (+0.12/-0.10) \times 10^{-10} \text{ m/s}^2$ by $z \sim 1$, excluding a constant a_0 decisively, killing the MOND picture ED contradicts and matching ED’s local value to $\sim 8\%$. One tension remains, on the rate, and it is the survey’s *own* headline conclusion: it fits a *linear* $a_0(z)$ and reports the evolution running **faster than $H(z)$** — which is directly the anti-ED direction, since ED is forced to exactly $H(z)$. Recasting the survey’s measured points as a power law gives $\alpha \approx 1.18$ — an ED-side re-fit, *not*

a figure MUSE-DARK III itself reports — which places ED’s forced $\alpha = 1$ a nominal $\sim 4\sigma$ from that recast (its ± 0.04 an underestimate), softening to $\sim 1-2\sigma$ once one-survey systematics are folded in; MOND’s $\alpha = 0$ is dead regardless. So ED’s qualitative call is confirmed and MOND’s excluded, while ED’s exact rate is disfavored but not refuted. The decisive test is a direct, multi-survey $a_0 \propto H(z)^\alpha$ fit; $\alpha = 0$ is already dead, and ED lives or dies on $\alpha = 1$. This is the most distinctive, most live bet in the document.

1. **The participation-envelope ratio.** ED predicts a ratio $f_v \approx 1.1 \cdot v_{\text{dec}}$ (the Q/π form) between a slow coherence-envelope modulation and the decoherence rate. Its claimed universality rests on the oscillator quality factor $Q \approx 3.5$ being universal, which is assumed rather than derived; the coefficient was itself corrected sevenfold (from a stale $8 \cdot v_{\text{dec}}$) shortly before this writing, and the spatial-PDE harmonic cross-check is so far inconclusive. It is the boldest of the bets and the least settled, offered as a distinctive test, not an established ratio. No standard theory predicts a universal ratio spanning optomechanics to biological FRAP; decoherence is supposed to be monotonic with no slow envelope. The detector is built and validated; the accessible band is $\sim 11-110$ Hz (a FRAP Lomb–Scargle measurement). Commissioned FRAP is currently unfunded, so the live zero-cost paths are a literature-data dig or the spatial-PDE harmonic legs.
2. **The merging-cluster knee.** ED predicts a sharp, step-like knee in the offset-velocity relation of merging clusters (flat $\rightarrow$ knee $\rightarrow$ linear $\rightarrow$ ceiling), with fast mergers showing more than two lensing peaks. Λ CDM predicts velocity-*independent* scatter (no knee); MOND predicts a *smooth* roll-off (no knee). The shape is robust; the knee’s location is inherited. A closer look tempers the “testable now” billing: assembling the current dissociative-merger sample (an 11-cluster compilation of published mass-gas offsets and merger velocities) shows the ceiling-and-multiplicity leg is only *weakly* checkable today — every catalogued merger sits below the saturation velocity so the ceiling is unstressed, and the multi-peak systems are complex multi-way mergers rather than the fastest, so multiplicity is confounded. The heterogeneous literature offsets scatter by an order of magnitude against the advection law. So the sharp knee is a *survey-era* test (Rubin, Euclid, eROSITA, plus a uniform re-measurement pipeline), not a near-term one; the current data neither contradict nor confirm it.
3. **The Class-A decoherence wall.** ED predicts a sharp, architecture-*independent* mass wall for quantum superposition at 140–250 kDa, with a few-percent second-harmonic fringe near the crossing. Standard decoherence is smooth and environment-dependent, with no universal wall. Matter-wave interferometry is at ~ 25 kDa and climbing.
4. **The error-correction plateau.** ED predicts logical error floors that do not go to zero ($\Gamma_{\text{plateau}} > 0$), where standard fault tolerance predicts error $\rightarrow 0$ with redundancy. Two developments since the inventory was ranked, the second stated plainly rather than smoothed. First, the plateau’s budget is not a free-standing postulate: it is the *same* finite V5 cross-chain budget that sets the entanglement-monogamy bound and the black-hole Page-curve budget, one bounded envelope seen in three arenas, tied in a fixed ratio of $1 : 1 : O(1)$ (the cross-chain and monogamy budgets identified at the substrate grain; the entanglement/area-law factor a measured ~ 0.88 , one of several convergent $O(1)$ counts). Second, a **do-main restriction made under tension, disclosed as such:** the original Class-C paper predicted that *surface codes* plateau in code distance and made that the near-term test. The leading surface-code demonstrations instead show clean exponential suppression with no plateau in the tested range, and the claim was subsequently narrowed — the mechanism is now argued to bind only on *broadcast-type* redundancy (repetition codes, cat states), not on topological surface-code distance, where the per-site budget load stays bounded. The narrowing rests on a legitimate fact (topological codes keep correlations local), but it was made in response to that tension, and the follow-up paper’s claim that the original had “anticipated” it overstates the original text, which predicted the opposite. Honest status: surface-code suppression is *now* consistent with ED by construction, not a passed prediction; the live test has moved to a *pair* on public data — the persistence of the repetition-code error floor (an apparent $\sim 10^{-10}$ floor in long runs, attribution to correlated bursts still contested, a watch item not a confirmation) and a ceiling on certifiable cat-state width. The distinctive claim is the shared budget and its ratios; the near-term test is whether the broadcast floor survives mitigation at cross-platform-consistent content.
5. **BTFR zero intrinsic scatter.** ED predicts *exactly zero* intrinsic scatter in the deep-MOND asymptote of the baryonic Tully–Fisher relation, and slope exactly 4. Both are consistent with SPARC (slope $3.85 \pm$

0.09, near-zero intrinsic scatter). But a SPARC confrontation shows this is *consilience, not an ED-vs-MOND weapon*: slope-4 and zero-scatter are shared with MOND, so they discriminate ED-and-MOND from Λ CDM (whose halo models need ≥ 0.15 dex intrinsic scatter), not ED from MOND. The one ED-distinctive piece is the normalization $a_0 = cH_0/(2\pi)$, which MOND fits and ED derives, and that is a $\sim 10\%$ postdiction (matched within the RAR systematic), not a forward win. The genuinely forward, ED-vs-MOND edge of the BTFR cluster is the redshift evolution of a_0 , the flagship above. So BTFR stays a strong Λ CDM-tension consilience that ED passes, and is demoted here from a headline weapon.

6. **Dark energy pinned to $w = -1$.** ED predicts saturation, not quintessence: the dark-energy equation of state is pinned to $w = -1$ with no evolution. This is structurally forced, not a free choice: a horizon-tied vacuum energy tracking the *live* expansion rate would give $w = 0$ and no acceleration (§5), so ED's saturation vacuum must sit at exactly $w = -1$, and it structurally cannot cross the phantom divide ($w < -1$) either. Quintessence and phantom models allow $w \neq -1$ and $w_a \neq 0$. DESI and Euclid are measuring this now; if the hint of evolution firms up, it is a near-term kill — which is exactly what makes it a real weapon.
7. **$\alpha_2 = 0$ exactly, and the matter-sector prohibitions.** The preferred-frame parameter α_2 is identically zero (§5), where a generic khronometric or Lorentz-violating theory gives $\alpha_2 \sim O(1)$; this is a kill-switch ED passes (GR passes it too, so it is a survival, not a win over GR). And the matter sector carries hard prohibitions from the clean-vector theorem (§10): no parity-violating chiral $U(1)$ force, no stable gauge group beyond $SU(3)$, no anyons in genuine 3+1D. Any counterexample is a one-line kill, and competitors do not make these prohibitions.
8. **Non-thermal Hawking radiation and a Planck-mass remnant.** ED predicts a sign-definite non-thermal correction to the Hawking spectrum, $n_H[1 - c_1(\omega/\omega_c)^2]$, and that evaporation halts at a stable Planck-mass remnant rather than running to zero. Standard semiclassical gravity gives an exactly thermal spectrum and complete evaporation. Analog-gravity systems (BEC, optical) are the near-term test; the cosmological abundance of Planck-mass remnants is a longer-term one (§6).

Several of these trace directly to arrow-sector results: $\alpha_2 = 0$ and the pinned dark energy come from the khronon (§5), and the matter-sector prohibitions come from the clean-vector theorem (§10). The arrow that unifies the report is also where several of its sharpest falsifiers come from.

The confirmed floor

Two predictions are settled, and it is important to state exactly what they are worth.

- **The Universal Degenerate-Mobility law** $D(c) = D_0(1 - c/c_{\max})^\beta$. ED forward-predicts the *functional shape*; the exponents ($\beta, c_{\max}$) are fit per material (Paper_085: “matching, not derivation”; population $\beta = 1.72 \pm 0.37$), and the shape then holds across ten materials at $R^2 > 0.986$. So it is a confirmed *shape*, not a confirmed number — the corpus's one genuine confirmed forward prediction at that tier, and the floor the report stands on.
- **$c_{\text{GW}} = c$** (GW170817, to $< 10^{-15}$) is what killed many rival modified-gravity theories. But ED inherits it structurally (§5), so it is consilience, not a distinctive win.

Neither of these is the argument-ending weapon. The mobility law is a soft-matter scaling law that other frameworks can accommodate; the GW speed is a survival ED shares with GR. They establish that ED makes correct forward predictions; they do not, by themselves, single ED out.

The north-star

The plain status: **no distinctive, argument-ending prediction is confirmed**. One of them now has distinctive data in hand, and it is worth stating exactly what that partial result is. The flagship, a_0 's redshift evolution, has met its first survey: ED's qualitative call is confirmed, a_0 evolves and a constant a_0 (MOND's) is excluded decisively, with ED's local value matched to $\sim 8\%$; but ED's *exact* rate ($\alpha = 1$) is mildly disfavored (a nominal $\sim 4\sigma$, softening to $\sim 1\text{--}2\sigma$ under one-survey systematics), the data preferring a somewhat faster evolution. That is a genuine, live, distinctive confrontation, not a confirmation, and not a refutation. The other bets, the envelope ratio, the cluster knee, the Class-A wall, remain unconfirmed. And the recent foundational work is clear about

its own limits here: the quantum-logic reconstruction (§4) and the chiral-gauge sector (§10) closed structure and located walls, but they produced *no new near-term weapon*. So the highest-leverage move toward the goal is not further derivation; it is pressing the bets above, the a_0 -evolution fit first.

Two distinctive claims are explicitly *not yet* weapons, and this report flags them as such rather than inflating them. Primitive einselection and the emergent multi-context structure (§4) are genuinely distinctive but have no concrete near-term observable. And the gauge-handedness/matter-sign correlation (§10) is theoretically firm but is flagged in the register as not testable with current data, so it belongs here as a distinctive-but-not-yet-testable claim, not a live weapon.

What this buys

This is where the report is most exposed, and deliberately so. ED is falsifiable at many sharp, distinctive points — a universal envelope ratio, a step-like cluster knee, an architecture-independent decoherence wall, an exactly-zero BTFR scatter, a pinned dark-energy equation of state, a set of one-line matter-sector prohibitions. Its credibility as a unified framework ultimately does not rest on the derivations in §3–§14. It rests on confirming one of these, and that has not happened yet. The report’s final section states what that means.

Related papers.

- *Falsification Register and Prediction Inventory* (predictions/Paper_101_FalsificationRegister);
- *The Class-A Wall*
- *Baryonic Tully–Fisher*
- *Offset–Velocity Law*, the merging-cluster test
- *Hawking Spectrum*

§16 — The Bottom Line

Event Density is **form-complete and not closed**, and both halves of that sentence are meant precisely.

What it is

ED classifies the *form* of every sector a unified framework must address, and it does so from one primitive. The arrow of commitment reconstructs quantum mechanics by selecting the pointer basis and supplying measurement repeatability (§4); it closes gravity as the khronon, a genuine preferred time, unifying the weak-field Einstein metric with dark matter and dark energy in one field (§5); and, most distinctively, those two jobs are the *same* arrow, so the problem of time dissolves rather than being reconciled (§6). Around that spine, the Standard Model sectors are the same primitive in smaller roles: it quantizes and protects charge and casts electromagnetism as a coarse-grained shadow (§9), it is half of the chirality operator and forces parity violation to be spontaneous (§10), it undoubles the spinor and makes a lone excitation massless so that mass comes from binding (§11), and it anchors the single hypercharge of the derived gauge structure (§8). The values that fill in these forms — the couplings, the masses, the mixing angles, the cosmological constant’s magnitude (which reduces to H_0) — are inherited, and inherited *by principle*: a pure-relation substrate produces no intrinsic scalar, so it cannot and need not set a dimensionless number (§12), exactly as the Standard Model inherits the same twenty parameters. (Dark energy’s *structure* is accounted for and the naïve mode-tower catastrophe behind the 122-order problem dissolved — §5, §12; it is the value, the last measured scale H_0 , that is inherited.)

That is a serious and coherent claim. One primitive, worked independently across sectors that mainstream physics treats as unrelated, keeps turning up as the mechanism, most strikingly at the black hole, where the arrow’s irreversibility preserves information as a record and addresses the information paradox (§6).

What is open

Two structural questions are genuinely unbuilt. Why the gauge multiplicities are $\{1,2,3\}$ (#1, §8) and whether ED’s exact charge conservation forces anomaly-freedom on the emergent chiral content (#2, §13). These are not

unexamined, and they share a root: the substrate-to-Dirac worldline reduction, which both gate on, is itself form-complete (operator form forced, continuum limit computed, undoubling verified, §11). What remain are two residuals of that one reduction — #1’s rep-spectrum reduces to the graph-native-spinor / channel-topology wall, while #2’s anomaly face gates on a *sibling* residual (the non-Hermitian→unitary / covariant-worldline piece) not shown to be the same wall — plus the rep-spectrum’s conjectured linking route. So the two opens **converge on one reduction without collapsing to a single wall**: one root, two distinct residuals, one conjectured route. Both are unsolved in the Standard Model too, which postulates what ED is trying to derive.

One wall is proven: the finite-memory substrate provably cannot reach the primality escape layer (§14), theorem-anchored, the program’s one certified negative. And the remaining limits — three-dimensionality, the chirality casting, the constants — are principled inheritances, not open derivations.

What is not done

Most importantly: **no distinctive, argument-ending prediction is confirmed** (§15). ED makes one confirmed forward prediction (the *shape* of the degenerate-mobility law; its exponents are fit) and survives the tests that killed rival theories ($c_{GW} = c$). Its flagship distinctive bet, the redshift evolution of the MOND scale $a_0(z) = cH(z)/(2\pi)$, has now met its first data and returned a *partial* result: the evolution is real and a constant a_0 is excluded decisively, so ED’s qualitative call is confirmed and MOND’s is killed, but the observed rate is somewhat faster than ED’s mechanism-forced $\alpha = 1$, a mild tension (nominal $\sim 4\sigma$, $\sim 1-2\sigma$ under systematics) from a single first-generation survey. That is a live, distinctive confrontation, not a confirmation. The other bets that would single ED out cleanly — the universal envelope ratio, the step-like cluster knee, the architecture-independent decoherence wall — are still open. Until one of these confirms, ED is a coherent, falsifiable unified framework, not a certified one. That distinction is the ceiling of this entire document.

What would close it

Three things, in rough order of decisiveness:

1. **A confirmed novel prediction.** The decisive one. Executing the near-term Tier-1 bets (§15) is the highest-leverage work toward it, and the corpus is explicit that further derivation is *not* — the recent foundational work closed structure but produced no new weapon.
2. **The substrate-to-Dirac reduction (T4).** This single arc gates the relativistic chirality statement, the graph-native spinor, and the one non-inherited anomaly candidate. Closing it would convert three of the matter sector’s “form-complete, structure-inherited” results into deeper derivations.
3. **The linking argument for the internal dimension.** Establishing that committed order is held by linking *specifically* — the one open premise behind the 3D route — would tie the internal dimension to three and un-wall #1. It is conjectured, not a delegatable computation, and not even askable on the certified 2D simulator, so it is the least near-term of the three.

Conclusion

ED is a unified framework — one primitive, the arrow of time, doing the load-bearing work across quantum mechanics, gravity, and the Standard Model’s structure — that is form-complete, value-inherited by its own logic, explicit about two genuine structural opens and one proven wall, and not yet confirmed at any of its distinctive falsifiers. The case it makes is that the recurrence of a single primitive across sectors is real and worth taking seriously; the case it does not yet make is the empirical one that would end the argument. This document carries both openly and states them at their true tier.

Related papers. The consolidated open/closed map is event-density/docs/ED_Open_Derivations_Ledger.md and ED_Research_Targets.md; the full corpus index is PAPERS_INDEX.md. Each section’s papers are listed at the end of that section.

Appendix A — The certified simulations

The “measured” tier means a result was produced by a built-and-run simulation of the certified substrate (or, for the analytic probes, a direct computation of the stated quantity). These are the runnable artifacts behind the report’s measured claims; paths are in the ED-Generative and event-density repos.

Result (§)	Script	What it runs
Charge skeleton, integral Gauss law (§9)	holonomy_test.py, coupling_test.py, sourcing_test.py, relaxation_test.py	B4: winding quantization to $\mathbb{Z}$; w-indexed bandwidth ladder; circulation = $2\pi w$ loop-independent; the ontology fork ($1/r^2$ only off-ED) — evaluation/B4_Arc/
Maxwell shadow (§9)	maxwell_from_coherence_probe.py	FFT Poisson solve of the coherence-action minimizer around a point charge; $p=1$ (3D Coulomb) fit $R^2=0.97$ — evaluation/B4_Arc/
QM orthogonality + covering law (§4)	move1_operational_orthogonality.py, gleason_nonboolean_probe.py	perfect distinguishability $\Leftrightarrow$ orthogonality ($1-c^2$); the non-Boolean complementarity gate — evaluation/ChiralGauge/
Chirality: clean-vector theorem (§10)	rep_spectrum_casting_winding.py	parity-clean point-gap winding = 0 for $N=1.6$ (control on the theorem); broken-case winding = N — evaluation/ChiralGauge/
Spinor undoubling (§11)	chiral_3p1d.py	Nielsen–Ninomiya: Hermitian naive = 16 doublers, arrow (Wilson term) = 1 survivor at the origin — evaluation/ChiralGauge/
Binding mass + inertia (§11)	mass_from_binding_probe.py	free front $v=0.98$; unbound vs V5-confined (extent 55 vs 1.4–2.3); COM sub-luminal; inertia 0.72 vs unbound-control 0.97; the k_{11} time-dilation discriminator — theory/Higgs_Emergence/
Khronon dynamical rule (§5)	GR-III simulation ($\ddot{b} = D\nabla^2 b - \kappa\rho$)	Newtonian fixed point; $r_s \propto M$; frozen $b \rightarrow 0$ horizon + area law; khronon speed $c_s = c$ — gravity arc
Channel capacity = 0 (§6, §12, §14)	A1 channel-capacity probe	controlled capacity exactly zero (the internal reading of the finite-reach ceiling)
Finite-memory ceiling (§14)	primes_test.py	sieve to $N=5 \times 10^6$: template reproduced (1.700-bit invariant, lpf ladder); escape blocked (optimal Möbius correlator $\rightarrow 0$; twin density needs $2C_2$) — evaluation/Primes_Arc/

Verification status: every script path above is confirmed present. Four reproduce their cited numbers exactly on re-run — rep_spectrum_casting_winding.py (clean winding 0 $\forall N$, broken = N), chiral_3p1d.py (16

doublers $\rightarrow$ 1 survivor at the origin), `maxwell_from_coherence_probe.py` ($p=1$ 3D-Coulomb, $R^2=0.97$), and `mass_from_binding_probe.py` (inertia 0.72 vs unbound-control 0.97). The remainder (the B4 holonomy suite, `move1/gLeason`, `GR-III`, `A1`, `primes_test.py`) carry the numbers reported in their source papers.

Scope note: the gauge sector's non-abelian action and mass-gap (§8) are analytic (gauge-program tier), not certified-simulator runs — only the abelian/Maxwell case is simulator-grounded. The chirality theorem's winding computation is a control on a proof, not an independent measurement. Both are flagged as such in their sections.

Appendix B — A worked derivation: charge quantization from the substrate

This appendix carries one representative result from the primitives to the conclusion with no citation, so a reader can check the report's *method* on a self-contained example rather than trust a reference. The charge skeleton (§9) is the cleanest case; nothing beyond the primitives named below is used.

Inputs. Three primitives and one structural fact:

- **P09** (U(1) polarity): each participation at a node carries a phase $\phi \in \mathbb{R}/2\pi\mathbb{Z}$ — an angle valued on the circle S^1 .
- **P05** (polarity-transport): a discrete U(1) connection on the participation graph; transporting the phase across an edge e advances it by a real increment $\Delta\phi_e$ (the link variable).
- **P11** (commitment irreversibility): a committed node's polarity is single-valued (one definite angle, mod 2π), and a commitment cannot be undone.
- The participation graph carries cycles (independent closed loops).

Step 1 — the holonomy around a loop. For a closed loop $L = (u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_n = u_0)$, transport the phase around it. The accumulated phase — the holonomy — is the sum of the edge increments: $H(L) = \sum_{e \in L} \Delta\phi_e$.

Step 2 — single-valuedness quantizes it. Transport returns to the start u_0 . By P11 the polarity there is single-valued, so the transported phase $\phi(u_0) + H(L)$ must equal the committed value $\phi(u_0)$ as an angle, i.e. modulo 2π . Hence $H(L) \equiv 0 \pmod{2\pi}$, so

$$H(L) = \sum_{e \in L} \Delta\phi_e = 2\pi \cdot w, \text{ with } w \in \mathbb{Z}.$$

The integer w counts how many times the transported phase wraps the circle around the loop. This is forced, not fitted: the connection's target is a circle, and the winding of a loop into a circle is $\pi_1(S^1) = \pi_1(U(1)) = \mathbb{Z}$. The quantization is a fact about the topology, independent of the individual edge values.

Step 3 — conserved and irreversibility-protected. w is locked. The edge increments $\Delta\phi_e$ may vary continuously, but a continuous change cannot move an integer: a real sum constrained to lie in $2\pi\mathbb{Z}$ can only jump between integers discontinuously. The one way to change w is to un-set and re-set a committed polarity — an un-commitment — which P11 forbids. So w is a conserved topological invariant, protected by irreversibility.

Step 4 — the integral Gauss law. Let L and L' be two loops that enclose the same winding source and nothing else between them. Their composite boundary $L - L'$ encloses no winding, so by Step 2 its holonomy is $2\pi \cdot 0 = 0$, giving $H(L) = H(L')$. Therefore the circulation $\sum_{e \in L} \Delta\phi_e = 2\pi \cdot w$ is the *same* integer for every enclosing loop L , independent of its size and shape. This is the integral content of Gauss's law: the flux through any surface enclosing the source is fixed by the single integer w — unscreenable and topological.

Result, and honest scope. From P09 + P05 + P11 on a cyclic graph: a conserved, irreversibility-protected integer winding $w \in \mathbb{Z}$, sourcing a field through a loop-independent circulation $2\pi \cdot w$. That is the topological skeleton of charge — quantization, conservation, and the integral Gauss law — reached graph-first, before any comparison to charge. What it does *not* give, and does not claim: a charge magnitude or spectrum (w carries no selected value; the $\pm 1, \mp 1/3$ pattern is not produced), a determined local $1/r^2$ field (the circulation is fixed, but its per-edge distribution is undetermined), or the identification of w with electric charge. The derivation is of the *structure*, at exactly the tier §9 assigns it.